AI Virtual Makeup Try-On

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DECLARATION

We hereby declare that this project report entitled "**AI Virtual Makeup Try-On**" is written by us and is our own effort and that no part has been copied or taken without a mentioning reference of source.

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CERTIFICATION

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CHAPTER 1

INTRODUCTION

With the changing consumer needs come new digital tools and that changing consumer needs leads to the continuous evolution of the makeup industry. Today consumers especially women have formally begun to rely on online shopping platforms to purchase cosmetics. But the absence of the sensory in store experience where customers can physically try out the makeup makes it difficult for them to ascertain on how the products will appear on their skin. If picking out products for a special occasion wedding, photoshoot or displaying them to the public, this holds especially true [1].

Often, in store testing of makeup involves physical testers which can be unhygienic and inconvenient, and online shopping for makeup lack the visual feedback to see how a product will look on one's skin [2]. One emergent solution to the challenge is virtual makeup try-ons that enable customers to try on products virtually before purchasing. Beyond high adoption rates for such technologies, there is a great potential to fill the practice gaps of real time customization, product accuracy and personalization to create a more exciting 'next' one for the users [4, 5].

The aim of this project is to enable the creation of a virtual, real time, AI based virtual makeup try on tool, which can simulate the makeup application on a person's face, so the makeup products will look in advance of actual application. It will bring convenience, personalized, and efficient experience, a virtual makeup mirror for consumers to apply makeup at the home, and salons for clients (esp brides), especially brides require a perfect look on their special day.

1.1 PROBLEM STATEMENT

The current process of choosing makeup products is an ongoing problem that many customers, mainly women, face in order to achieve their perfect makeup look in a very short time and also without much effort. The following are the key problems based on the following points:

1. **Lack of Visualization:** Users can't see how makeup will actually look on their face before applying it. Foundation shades and lip colors often appear different on each individual, making selection difficult.
2. **Hygiene Issues in Stores**: Using shared testers at makeup counters poses hygiene risks, particularly for brides and during busy events.
3. **Online Shopping Limitations**: Online buyers can't try products before purchase, leading to confusion and potential dissatisfaction—especially when it comes to shades.
4. **No Real-Time Customization**: Most existing virtual tools use static images and don't support real-time facial tracking, limiting interactivity.
5. **Lack of Personalization**: Users have unique skin tones and face shapes, but most tools don't offer product suggestions tailored to individual features [3].
6. **Bridal Makeup Stress**: Brides go through multiple trials to finalize their look, often under time pressure and with uncertainty about the results [6].

In an effort to solve these problems, this project aims to develop a real time virtual makeup try on system with the help of AI and AR technology where a user can apply makeup on their own face at home or in a salon setting in realistic and interactive manner [3].

1.2 PURPOSE OF THE APPLICATION

The AI Based Virtual Makeup Try On application aims to serve as a solution for makeup fantasy, especially women and brides, to view a makeup product virtually, so that the product can be tried on virtually in advance before applying it. The app will create an augmented reality experience via incorporating Augmented Reality, real time facial recognition and image rendering to show the user exactly how a particular makeup product looks on their skin.

The key goals of the application are:

- **Real-Time Virtual Makeup Application**: The app shall virtually apply makeup on the face of the user using the AI based facial recognition along with AR technologies. The users will be able to use lipstick, foundation, blush and eyeliner to paint themselves, applying the makeup and can see themselves immediately.

- **Personalized Makeup Experience**: The system will suggest makeup products best suited for user's facial features and skin tones based on the facial features and skin tones. This will aid the customer to make better buying decisions, especially when it comes to picking shades of foundation, lipstick or eyeshadow.
- **Convenient Virtual Makeup Trials**: Users will be able to try out different makeup and manipulate each for a varying intensity or coverage to see the perfect look for their skin tone and face. Such convenience can enhance the ability of consumers to make the right decision without having to go to a shop better to experience or taste several products.
- **Bridal Makeup Assistance**: This application will include the application processes for the brides, giving them an opportunity to try on wedding makeup looks from the comfort of their homes. It reduces the time on face trial, and is an interactive way of planning their big day look.
- **Retailer Support and Sales Enhancement**: The application will be designed to be integrated with the online retailers to enable customers to purchase the makeup products that they virtually try on. This smooth one from virtual trial to topping up for sale could decrease product returns and increase sales conversion rates.

1.3 PROJECT OBJECTIVES

The AI-Based Virtual Makeup Try-On project will focus on achieving the following objectives:

1. To develop a real-time makeup simulation system that virtually applies makeup on the user's face using facial landmark detection and augmented reality (AR) technologies.
2. To allow users to adjust makeup intensity and coverage in order to find the most suitable look for their skin tone, thereby improving decision-making and reducing reliance on physical product trials.
3. To implement high-quality graphics rendering techniques that ensure realistic blending of virtual makeup with the user's facial features under natural lighting conditions.
4. To provide a dedicated bridal makeup trial feature that enables brides to preview and finalize their wedding day look with ease and accuracy.

5. To enable future integration with retail and e-commerce platforms so users can directly purchase the makeup products they try within the application.
6. To enhance customer engagement by offering a fun, interactive, and personalized virtual makeup experience that encourages repeat usage and product exploration.

1.4 PROJECT SCOPE

For the AI Based Virtual Makeup Try On application, its scope is to provide an enhanced virtual makeup experience to users that will include the users to try the makeup and test different makeup products. It will be designed as a desktop based application in the first place and will later be expandable to mobile platforms.

1.4.1 MAKEUP PRODUCTS SUPPORTED

The supported key makeup products in the application are:

- **Eyebrows**: App allows user to explore various styles of eyebrow shape and color, dark brow look or natural brow or bold brow look.
- **Eyeliner**: The styles include: liquid, Gel, pencil with the option of changing the thickness, color and style.
- **Eyeshadow**: Users get to play around with eyeshadow colors and styles in a range of neutrals to bright to their discretion for opacity and excellence.
- **Foundation**: The app will provide many foundation shades to match different skin tones. Based on user preferences, it will adjust the intensity of coverage to full to light.
- **Lipstick**: Users can test different lipstick shades (light, medium, dark), finishes (matte, gloss, satin) and everyone intensities — since they develop and apply lipstick every day.

All these features will give the users an opportunity to try to a large variety of makeup products to plag a personalized, total makeup.

1.4.2 REAL-TIME MAKEUP SIMULATION

Augmented Reality, and Facial recognition technology will be used in the application to put the makeup products in real time. The system will automatically apply a makeup product on users face after selecting based on AI powered facial detection, taking in to

view the users facial feature as well as their skin tone. Users will also be able to alter the intensity of each makeup product (such as the opacity of lipstick, or the thickness of eyeliner) to the desired level.

1.4.3 PERSONALIZED MAKEUP RECOMMENDATIONS

The application will provide personal recommendations of makeup using AI based on the user's facial structure, skin tone, and makeup tastes. For example:

- The system will have eyebrow shapes and eyebrow colors that are related to the face shape of the user.
- In addition, it will suggest its foundation shades of color which can be paired with the user skin tone to make an ideal finish.
- The makeup styles like lipstick, mascara, blush and eyeshadow shades will be customized for the user's preferences and in accordance with the user's makeup inclination.

1.4.4 USER-FRIENDLY INTERFACE (UI)

An intuitive and easy to navigate interface will be implemented in the application. Users will be able to:

- Change the application intensity of the makeup (such as lip colour depth, mascara volume, blush opacity).
- Change the application intensity of the makeup (such as lip colour depth, mascara volume, blush opacity).
- You can also preview different combination of makeup products for creating unique looks.

Users of the system will be able to experiment with different products in an easy way and see the effects immediately since the system will guarantee that makeup is applied smoothly and realistically.

1.4.5 RETAILER AND SALON INTEGRATION

The first version will be more one for the personal consumers, but future versions could integrate with the retailers and beauty salons:

- **E-commerce Integration**: Retailers can also integrate their makeup product catalog into the app so users can buy the products they try. This provides seamless shopping experience, users can test makeup virtually and purchase on the same product online.
- **Beauty Salon Integration**: It's a Makeup Mirror for beauty salons and makeup artists that take clients through a clean makeup makeup before applying. For brides or clients that will be attending special events, this is a great feature as they can see a few makeup looks and then choose the final look digitally.

1.4.6 Non E-commerce Features

The initial version will not have payment integration, but users will still be able to try on products, save their favorite looks, and forward the beauty consultants or retail professionals off-line to make purchases. It will make makeup trials a fun and fairly interactive thing without actual testers of the product.

1.4.7 Device and Platform Support

Desktop platforms (Windows and macOS) will be the first stage of development. While this has not been a mobile product so far, there is a strong potential for future versions to become mobile apps (iOS, Android). With an app mobile version, the app would afford users an opportunity to try on makeup whenever and wherever they wanted thus expanding the reach of the app.

1.5 Proposed Solution

To solve this, we develop a real time virtual makeup try on technology based on AI, AR and Advance image processing technologies. We are going to build the app that will give the users an opportunity to try different makeup products in real time, and show what makeup products will look like if they applied on their face.

1.5.1 Key Technologies Used

- **MediaPipe**: For real time facial recognition and facial landmark detection, this powerful framework will be used, MediaPipe. The facial landmarks from MediaPipe's face mesh model will be precise enough to accurately locate the boundaries of makeup.

- **Augmented Reality (AR)**: An AR technology will help the makeup products to apply directly to the user's face in a natural way and can be adjusted while moving the face.
- **Image Processing & Graphics Rendering**: These techniques will ensure makeup rendering products with high quality effects into consideration of user's features like skin tone, lighting condition and make realistic simulation.

1.5.2 FEATURES OF THE APPLICATION

It will have the following features in the application:

- **Real-Time Makeup Application**: Real time application of the makeup products like lipstick, eyeshadow, foundation, blush and mascara.
- **Eyebrow Customization**: People can try out various eyebrow styles as well as eyebrow colors, determining thickness and opacity.
- **Eyeliner** : With the Eyeliner, users can play with various eyeliner styles and mascara volumes and change thickness and intensity for dramatic effects.
- **Personalized Recommendations**: Based on the user's face shape, skin tone, as well as style preferences, the user will get AI driven makeup recommendations.
- **Customizable Intensity**: Users will be able to customise each makeup product in real time to the level you want (e.g. lipstick shade, foundation coverage and eyeshadow opacity).

1.5.3 USE CASE SCENARIOS

1. **At-Home Users**: The app can show a woman how different shades of foundation will look on her face. Then, she can adjust the foundation coverage, the shade that aligns with her skin nicely.
2. **Bridal Makeup Trials**: The app allows the bride to have a trial of the different wedding makeup looks; classic style to bold styles, so she would look perfect on her big day.
3. **Beauty Salon Clients**: A makeup artist in a salon can use the app to assist a client in choosing a makeup style or a number of options for lipstick, foundation and blush.

1.6 PROJECT SIGNIFICANCE

There are many benefits that this project could yield for both consumers and retailers:

- **For Consumers**: Mainly it is for the Consumers as the application provides a hygienic, personalized and interactive ways to try makeup products. With this, users can experiment with the makeup using their own pace and select the appropriate products for their skin tone, facial features and their preferences. For the brides, the app is here to serve as a stress-free solution for them to choose the best wedding makeup.
- **For Retailers**: It will be an innovative sales tool for Retailers. Retailers are able to boost consumer confidence as well as conversion rates with a virtual makeup trial experience. Being able to e-commerce integrate with the makeup products would make it easier on the users and will simplify it for them to purchase the makeup products they tried of.
- **For Beauty Salons**: This app can be utilized by salons to provide virtual makeup consultations to the clients without actually having to visit the salon to attempt any differences. It will avoid spending time and ensure more precision results, especially in cases like weddings.

This project offers to change the paradigm in the way customers shop for makeup and prepare their beauty looks as it provides a real, personalized, and interactive makeup experience, making the whole process faster, easier and more pleasant.

1.7 CHAPTER SUMMARY

This chapter talked about the important elements of the AI Based Virtual Makeup project try. Talking about the biggest problems facing the consumers when it comes to the makeup products inability to see the look of makeup before it is used. In that regard, the proposed solution, using the AI, AR, and image processing, is introduced that can offer a real-time virtual makeup try-on tool for the users to apply at least various makeup products on their faces. This application will present to the users a personalized, interactive, and realistic makeup application, which will lead to higher consumer satisfaction with greater retailer sales.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION TO VIRTUAL MAKEUP TECHNOLOGY

The beauty and cosmetics industry has in fact turned virtual makeup try-ons into an essential need. Consumers often have to test products in store, using the traditional makeup purchase process, and there are inherent risks with hygiene, time constraint and not being able to really tell how it will look before application. These days, as online shopping and e-commerce has risen to such huge popularity, these limitations have actually grown even more problematic, and it is now that virtual makeup technology may possibly solve these problems [12, 10].

Such virtual makeup tools harness today's hottest Augmented Reality (AR), Artificial Intelligence (AI) and Computer Vision technologies to let users conduct virtual makeup product trials. These tools help you out with the process of applying makeup on the face of a user so that they can see how the makeup will be applied on your face in real time before you purchase. This section will discuss the history and development of virtual makeup technology, the market competition, and the technologies involved in the process [11, 9].

2.2 EVOLUTION OF VIRTUAL MAKEUP TOOLS

Regular structures of virtual makeup technology went from simple; makeup overlay on stills to later real time AR based simulation. The key stages in the development of the virtual makeup tools will be described in this section:

2.2.1 EARLY ATTEMPTS AT VIRTUAL MAKEUP

Virtual makeup applications at their initial stage operated using static digital images as the foundation. Users uploaded photos to have the software analyze facial points to virtualize several cosmetic items starting from lipstick through eyeshadow until foundation. These tools had several limitations [8]:

- **Lack of Real-Time Interaction**: Users experienced major limitations since they had no ability to modify their virtual makeup after application and received static non-dynamic results.
- **Static Product Representation**: Users noticed that static product representations regularly provided poor color reflection because applied makeup items frequently demonstrated poor compatibility with users' facial characteristics as well as their skin tone.

The historical tools provided users with their initial experience of virtual makeup try-ons although they created fundamental groundwork for enhanced digital try-on solutions in the future.

2.2.2 The Advent of Augmented Reality

AR technology brought a revolutionary change to how people try virtual makeup through digital overlays [10]. Through AR technology users obtain immediate visual feedback of digital makeup applied to their faces by using their smartphones or webcam. Users obtained interactive features in virtual makeup tools when this technology allowed them to apply makeup products during real time and modify their selections as needed [12].

During 2010s, when the AR based makeup tools took AR based makeup tools to AR based makeup tools like AR based makeup tools maker Sephora AR based makeup tools and L'ORÉAL Makeup Genius applications. Experimental capabilities were opened up for both instant and real time time consuming cosmetics product so users could try it out. New virtual makeup tools as a result of AR functionality enabled the additional addition of realistic interactive capabilities that morphed depending on the actual physical motion of the user.

2.2.3 Integration of AI and Personalization

During their development period, virtual makeup tools included Artificial Intelligence (AI) and had a partnership to offer better user interaction and individual product proposition. Skin color, as well as features of the face and some makeup preferences can be analyzed by the AI algorithms simultaneously and product recommendations can be provided that are as precise as possible. Continuous improvement of their recommendations based on

user interaction is enabled by the tools that are equipped with machine learning capability [9, 11].

When the market goes virtual, user preferences were considered with personalized suggestions that complement the given facial characteristics. The system then can make suggestion for skin tones and lipstick recommendations from previous makeup selections from AI technology foundation.

2.2.4 Real-Time Customization and Interactive Features

Users can also change the level of intensity on their makeup selections using real time capabilities. Users can apply lipstick tones from lighter to darker value through the use of virtual makeup tools, and they can alter foundation transparency from sheer to thick coverage and blush tones from subtle to vivid.

Real-time customization features are essential to users who wish to test makeup applications free of charge before buying real products. Users now achieve unprecedented flexibility because these tools enable them to try new looks which would have been impossible before now [8].

2.3 Key Players in Virtual Makeup Tools

A growing number of major businesses in the virtual makeup application market created innovative product trial tools for virtual makeup testing. These leading virtual makeup tools have different features available according to the review below:

2.3.1 Sephora Virtual Artist

The virtual makeup industry leader Sephora introduced its Virtual Artist platform to the market in 2015. Users access this tool to test different makeup products including lipstick and eyeshadow and foundation together with blush. Such AR-based platforms use facial recognition methods to pick out vital facial points for successful makeup application [4].

Key Features:

- Customers have access to various Sephora products which include beauty items along with skin care options and perfumes.

- This service enables foundation recommendations based on your skin tone.
- Real-time simulation enables users to instantly view the effects of different products through a virtual application process.
- Users can post their created looks to social media platforms so others can provide their opinion.

Limitations:

- The store presents solely Sephora-branded merchandise which reduces the selection of available products.
- The wide selection of products in the store is comprehensive yet the makeup application accuracy especially with foundation suffers when lighting conditions change.

2.3.2 L'ORÉAL MAKEUP GENIUS

Makeup Genius from L'Oréal introduced among the first major makeup brands to deliver AR-based real-time makeup testing functionality to customers. Through the app users get to experiment with various makeup items while developing fresh looks using their actual time video streams [6].

Key Features:

- Customers have access to an extensive choice of L'Oréal cosmetics including lipstick and foundation and mascara and other products.
- Real-time facial tracking for accurate makeup application.
- Users can apply products during live demonstrations to control the intensity levels for personalized outcomes.

Limitations:

- The L'Oréal Makeup Genius app enables users to try their makeup but it only works with L'Oréal products which reduces the overall product variety.
- Product suggestions tailored to skin color and facial characteristics are not available within the application's features.

2.3.3 YOUCAM MAKEUP

Today YouCam Makeup represents one of the top virtual makeup applications that users can find. Users can access a wide variety of makeup items encompassing lipsticks together with eyeshadows and blushes and mascaras within the application's selection. The software contains virtual hair styling functionality which supports its status as a complete beauty tool [5].

Key Features:

- It allows customers to find multiple makeup products between various business brands instead of sticking to one brand alone.
- Users can utilize the application to experiment with different hairstyles and colors of their hair.
- Users can apply makeup in real-time and modify either the intensity or color of their makeup through this system.
- The system determines individual recommendations using both user choice data and the distinct conditions of their skin color.

Limitations:

- The foundation matching accuracy needs further development since it remains challenged specifically with regard to matching skin tone correctly.
- VisualizeNG enables difficult use for people unfamiliar with makeup tools but particularly for novice users who learn its operation.

2.3.4 FENTY BEAUTY'S VIRTUAL TRY-ON

Through a virtual try-on program on Instagram AR Fenty Beauty made its diverse range of makeup accessible to users [7]. Through this tool users can virtually test numerous makeup products especially the well-known foundation shades.

Key Features:

- The platform enables customers to access and purchase the complete Fenty Beauty collection such as foundation as well as lip gloss and highlighters.
- Users can connect their looks to Instagram through an integrated feature that allows content sharing with their social media audience.

- The wide spectrum of foundation tones allows the product to accommodate various consumer skin complexions.

Limitations:

- Users face restrictions within the Fenty brand since they cannot try alternative brands or complete makeup looks through this platform.
- The app maintains space for improvement regarding the realistic simulation of application methods especially for eyeshadows and blush.

2.4 TECHNOLOGIES BEHIND VIRTUAL MAKEUP TOOLS

Virtual makeup effectiveness derives from multiple sophisticated technologies integrated into the system. Different advanced technologies are applied to virtual makeup applications which this section explains in detail.

2.4.1 COMPUTER VISION

The core function of virtual makeup tools depends on the use of Computer Vision technology [9]. The system detects facial features through detection protocols before tracking the landmarks for identifying key positions such as eyes and nose and lips and cheeks. Computer vision algorithms which come with video feed and single image capabilities allow precise accurate makeup placement of the user's face.

- **Facial Landmark Detection:** More than 468 facial landmarks are tracked by MediaPipe and Dlib deep learning models of facial landmark detection to ensure accuracy of cosmetic applications over facial features.

2.4.2 AUGMENTED REALITY (AR)

Close up the screen on a user's face via an AR experience, enabling users to see digitally applied makeup applied directly on their facial appearance in real time. As a result, this feature makes user experience appear highly realistic and interactive. Makeup products could be made dynamic real because AR technology permits the makeup product to adapt their position based on user facial movements throughout normal face expressions.

It consists of ARKit for iOS and ARCore for Android as an essential framework to perform real time interaction with facial feature in AR makeup applications. When secured using the tracking system each makeup product can be monitored to ensure the security of not only during smiling movement and conversation, but any facial expression.

2.4.3 Artificial Intelligence (AI)

Virtual makeup tools are made more personal with AI by suggesting products that are based on the user's face structure, skin tone, makeup preferences. As of natural face features as the core input, AI systems will discover the best makeup items for each individual users [13, 11].

- **Machine Learning Models**: AI models use vast makeup product and skin tone datasets to determine the best foundation shades together with appropriate lip colors and eye makeup selections.

2.5 Limitations in Current Virtual Makeup Tools

Virtual makeup tools have made substantial advancements yet multiple obstacles persist which reduce the tool's effectiveness and usability throughout its use. Virtual makeup applications will benefit from improvements to change the overall accuracy and flexibility and enhance their realistic appearance. Various limitations exist in the current virtual makeup solutions which we will outline in the following discussion:

2.5.1 Accuracy of Makeup Application

Virtual makeup applications face their biggest challenge because their solution does not match realistic makeup application effectively. Modern technological innovations fail to produce satisfactory makeup results that match reality. Several essential problems exist because of virtual makeup tools. Some of the key issues include:

- **Imperfect Blending**: Virtual makeup tools encounter challenges with blending foundations and blushes because these cosmetic products do not merge smoothly with customers' natural skin colors. A shortfall of foundation application occurs when it creates clearly visible lines at the border of real skin since it does not blend properly.

These issues become significant because the tool fails to adjust the product based on the user's skin texture and tone correctly [9].

- **Lighting Conditions**: The quality of real-world makeup application directly depends on how well the illumination settings match the current environment of use. The virtual tools have difficulty duplicating actual lighting since this limitation creates differences between what the digital try-on shows and how the makeup looks when applied in reality. Virtual makeup tools display products differently from how they look in actual light since they fail to match their appearance in natural and artificial conditions.
- **Inconsistent Product Application**: Digital face applications do not present a uniform product spread across makeup items such as lipstick eyeliner and mascara which becomes more evident when users express themselves. The user's facial moves during a smile cause lipstick to shift in appearance while mascara fades away from their eyelashes. Modern technology still faces obstacles when creating virtual applications and realistic solutions for responding to facial gestures.

2.5.2 LIMITED PRODUCT RANGE

The current virtual makeup applications provide users with limited makeup items in their catalogs thus they have constrained possibilities to explore different combinations. Common limitations include:

- **Limited Product Types**: The tools support only select makeup types which include mainly eyeshadows and lipsticks as their primary features [7]. Virtual makeup support ends at the most fundamental products including foundation in addition to mascara and blush since eyebrow products and contact lenses receive little to no aid. Virtual makeup tools become less useful to customers who want to try comprehensive makeup combinations involving multiple types of products.
- **Brand Limitations**: The number of brands customers can experience through virtual makeup solutions remains limited since programs function within either brand-developed applications or agreements with choose few cosmetics businesses. Users face restricted makeup exploration because they cannot test all available brands' products inside this platform. An ideal virtual makeup tool must enable users to test several makeup brands for them to discover optimal cosmetic products.

2.5.3 SKIN TONE MATCHING

The process of finding matchable skin tones for foundation products and other complexion products presents a major challenge in virtual makeup applications. Virtual makeup tools with AI and facial recognition technologies encounter multiple problems which prevent them from providing accurate services to users:

- **Detect Skin Undertones**: The perception of undertones in human skin differs among people so foundation color matching requires these undertones to create realistic results. The current technological tools experience significant difficulty when attempting to accurately match user skin tones particularly for dark and distinctive tanned consumers [13, 9].
- **Account for Environmental Lighting**: Environmental lighting plays a vital role in determining the appearance of makeup because it significantly affects finished results in physical makeup application. Modern virtual makeup tools do not recreate light reflection properly which results in unsatisfactory product presentation when users compare the results to their actual appearance in real life. The reflection of foundation shades varies significantly from one lighting scenario to another so they show inaccurate results under different conditions.

2.5.4 REAL-TIME CUSTOMIZATION

Virtual makeup tools on the market enable users to test different products but lack the ability to immediately change product intensity throughout their application process:

- **Product Intensity Adjustments**: Users desire to modify the intensity of their lipstick products to make them lighter or darker as well as control blush opacity levels. Current virtual makeup tools differ from professional makeup artists by lacking this fundamental capability of dynamic adjustment between makeup tones.
- **No Adjustment of Product Combinations**: Users desire to view combinations between different cosmetics especially when matching lipsticks with diverse eyeshadow choices. Users encounter limitations with product combinations in virtual makeup tools because they do not get full creative control during real-time application.

- **Realistic Texture and Finish**: Each makeup item including foundations requires customers to modify both its texture characteristics alongside its shine level and finishing quality. A user needs to check whether a matte or a glossy option of a specific lipstick product works better.

2.5.5 Facial Feature Adaptation

The availableExisting devices demonstrate limited capability to customize makeup products to match personal facial characteristics. For example:

- **Facial Shape Differences**: The shape of a person's facial structure determines the proper makeup application techniques because individual face shapes vary between oval round square and more. Users with round facial shapes need unique instructions for placing blush compared to those who have an oval face. The current generation of virtual cosmetics products lacks functionality which modifies makeup application methods to account for distinctive facial features that results in substandard or unreal output.
- **Facial Movement**: Virtual makeup products frequently fail to alter their appearance when users show facial expressions through smiling and blinking and changing expressions. The improper adherence of mascara and eyeshadow becomes a significant problem because the application tends to change after face movements. The system requires additional work to develop facial recognition technology which adjusts makeup appearance while users engage in facial movements.

2.6 Future Directions in Virtual Makeup Technology

The limitations described previously present multiple areas for potential improvement in virtual makeup technology development where new directions can be used for enhancement purposes:

2.6.1 Advanced AI and Machine Learning

AI-based virtual makeup will improve through better enhancements of algorithms to detect skin tones and personalize recommendations about products. AI will achieve further enhancement through these measures [13, 9]:

- **Better Skin Tone and Undertone Detection**: The latest AI models can detect more skin undertone variations to guarantee virtual makeup products specially foundation selections which align perfectly with individual skin tones.
- **Personalized Makeup Suggestions**: By analyzing user preferences, facial structure, and past product selections, AI can provide highly personalized makeup suggestions. Through training AI-driven systems gain knowledge about cosmetic matching for individual users to provide complete harmonious makeup selections for them.

2.6.2 Augmented Reality (AR) Improvements

The future development of augmented reality (AR) integrated makeup tools will achieve better realism in demonstrating makeup products. Key improvements include:

- **Enhanced Lighting Effects**: AR-based tools of the future will provide users with improved lighting simulations which replicate outdoor illumination beside indoor lighting thus giving users an exact picture of their makeup appearance in different lighting situations [8].
- **Improved Real-Time Interactivity**: The capability of AR to enhance interactivity should focus on adding features for live intensity adjustments as well as dynamic blending functions to offer users better virtual makeup control.

2.6.3 Real-Time Customization and Advanced Textures

Modern tools do not enable users to customize their makeup products in real time. Future advancements could include:

- **Customization of Texture and Finish**: Users should be able to modify the finish quality of their beauty products between gloss and matte textures or between satin and high gloss outcomes for foundation and lipstick types.
- **Full Makeup Combinations**: Users should be able to combine various complete makeup applications through the platform's functionality. A person can view a particular lipstick appearance that matches their chosen foundation along with blush combination.

2.6.4 FACIAL FEATURE ADAPTATION

Facial detection and tracking technologies at an advanced level will serve as requirements to develop realistic virtual makeup tools with better adaptability. Future innovation will enable more precise makeup application with realistic outcomes:

- **Facial Shape Customization**: Mathematical algorithms which consider different face shapes allow the system to apply makeup products precisely as they align naturally with individual facial structures.
- **Movement Tracking for Makeup**: Real-time tracking of facial movements through the app allows makeup to adapt automatically while users move without damaging their makeup appearance.

2.7 CHAPTER SUMMARY

Virtual makeup technology shows substantial developments yet various obstacles remain for its advancement. The current virtual makeup tools deal with several fundamental issues including imperfections in makeup placement as well as a narrow product selection and matching skin tones along with immediate customization of products. However, with advancements in AI, Augmented Reality (AR), and computer vision, these challenges can be addressed to create a more realistic, personalized, and interactive virtual makeup experience.

A subsequent chapter explains the project methodology that involves design choices and development steps alongside technology selection for constructing the proposed solution and its goal to surpass user experience obstacles.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

The development of the AI-Based Virtual Makeup Try-On application executes through multiple stages which fuse AI-based recommendations with real-time facial recognition while utilizing AR for an interactive and fluid makeup trial system for users. Our report describes the system development methodology through its development process and tools and technologies selection and testing strategies application [16].

Our development process relies on Agile methodology that delivers continuous development cycles through multiple testing sessions. The Agile sprint process concentrates on a single feature or functionality at a time which enables the system development to occur through gradual improvements stemming from regular user feedback. The chapter explains four essential phases which include facial detection alongside AI-based makeup suggestions followed by AR-based makeup application along with UI creation [14, 18].

3.2 DEVELOPMENT PROCESS

Before selecting a methodology for the development of our AI-Based Virtual Makeup Try-On application, we analyzed several popular software development models:

- **Waterfall Model**: A linear and sequential approach, where each phase (requirements, design, implementation, testing, deployment) must be completed before the next begins. While simple, it's inflexible and does not adapt well to changes during development.
- **V-Model**: An extension of the Waterfall model where testing is planned alongside each development phase. It's useful for projects with well-defined and unchanging requirements.

- **Spiral Model**: A risk-driven approach that combines elements of both iterative and Waterfall models. It is ideal for large-scale, high-risk projects but may be too complex for smaller applications.
- **Agile Methodology**: An iterative and incremental model that focuses on customer collaboration, adaptability, and frequent delivery of working software. It breaks the development into small sprints with feedback incorporated at every stage.

Why We Chose Agile

We selected the Agile methodology because it best suited the dynamic nature of our project, which involved:

- **Frequent Testing**: Since we had to verify real-time makeup application and facial tracking, Agile allowed testing after each sprint.
- **Iterative Improvements**: User feedback from testing sessions helped refine overlays, UI design, and performance optimization.
- **Flexibility**: Our requirements evolved during development (e.g., new product categories, slider adjustments), which Agile handled efficiently.
- **User-Centric Focus**: Agile's emphasis on continuous user involvement ensured a better end-user experience in terms of design, performance, and usability.

Each sprint in our development lifecycle focused on one key functionality—from facial detection to AI recommendations—allowing smooth integration and testing of each component before proceeding to the next.

3.2.1 Sprint 1: Facial Detection and Real-Time Tracking

Precise cosmetics outcomes are achieved by tracking faces first so that the detection system can start. In real time tracking, the facial recognition system has to perform two tasks for detecting the important facial features.

- **Facial Landmark Detection Using MediaPipe**: It uses the FaceMesh model on MediaPipe to find out 468 facial reference points in the entire face. Eye features then along with eyebrow are followed by facial landmark of the nasal region and mouth attributes, and concluded with chin identification. A real time detection accuracy of

these landmarks is essential as they will subsequently be used as basis points for makeup application in a later process [15, 17].

- **Real-Time Face Tracking**: The system will track the last moment (blink) and the motion trend (head turn) in real time and real eye and let the makeup match and keep the face made up. The makeup adjustments that will be made during user smiles will work automatically without affecting the initially applied makeup products. MediaPipe's face mesh is utilised for face tracking that preserves the stability makeup effects when the user's facial appearance changes [17].

3.2.2 SPRINT 2: MAKEUP APPLICATION AND AR INTEGRATION

This sprint continued with implementing Augmented Reality (AR) technology that actually allows for application of genuine makeup [16]. The facial landmarks will be detected and makeup products will be added on top of this face. While using AR, the platform keeps the resemblance to cosmetics as the cosmetic adapts to the changes of the person's face.

- **Real-Time Makeup Overlay**: Through Augmented Reality technology the system displays makeup items that include fingerpaint, blush, eye shadow as well as foundation all directly on the user's facial contour. The user's natural skin appearance guides the product placement to match their skin tone alongside facial shape and facial expressions. The foundation product adapts to match user skin tones precisely as it applies on the face while maintaining consistent shades with natural skin color [9].
- **Interactive Makeup Adjustments**: The system enables the user to modify makeup products through direct manipulation. Users will have the power to make structural changes to cosmetic elements such as modifying lipstick color tonality between lighter and darker values and managing foundation opacity along full-to-light spectrum and selecting blush intensity. A dynamic user-to-product interaction gives users complete control of their virtual makeup application.

3.2.3 SPRINT 3: AI-BASED MAKEUP RECOMMENDATIONS

During the third sprint phase the team develops the AI-driven recommendation system that gives users makeup proposals customized to match their facial characteristics and color type.

- **AI Algorithms for Skin Tone Detection**: An AI system performs skin tone analysis of user facial images to help users obtain suits matching their exact facial characteristics and unique skin complexion [9]. This includes:
    - Skin tone foundation shades according to user undertone classification (warm, cool or neutral) will be available for selection.
    - The system matches lipsticks and blushes to face types and selected color choices of each user.
- **Training the AI Model**: The AI model receives training through a database that contains multiple makeup products, skin tones, along with facial images. The AI system enhances its capability to generate exact and customized makeup recommendations through accumulating user data obtained from app interactions over time [13].

3.2.4 Sprint 4: User Interface (UI) Development

A user interface forms the basis for seamless interactions between users and an application system. The application will use React.js technology for its interface development because this framework provides real-time interactions and dynamic content updating abilities [18].

- **UI/UX Design**: The interface will present itself through a minimalistic layout which includes natural user interface navigation features. Users must easily reach all functions including selecting makeup items and adjusting them and real-time result viewing.
- **Real-Time Interaction**: Through its usage of the React.js framework users can have smooth real-time interactions with the system. Users can drag makeup items like lipstick and blush onto their facial representation so they instantly see how their appearance will look. Users can modify product intensities through the interface which shows instant updates on their changes.

3.2.5 Sprint 5: Testing and Quality Assurance

A complete cycle of testing procedures with quality assurance methods will be applied to the application during its final sprint phase.

- **Unit Testing**: Each distinct feature including facial detection and makeup application and recommendation system will subject to unit testing procedures to confirm their individual correct operation. Edge conditions involving different light conditions for

face detection and various skin tones for AI recommendation testing will be evaluated through specially written unit tests [14].

- **Integration Testing**: The testing phase of integration will follow unit testing to establish complete connectivity between system components. A test of the user interface must prove its ability to exchange signals with the AI system along with its capability to render proper makeup application from facial information.
- **User Acceptance Testing (UAT)**: Real users at UAT stage will test the virtual makeup application in actual scenarios to both verify accuracy and ensure ease of use. User feedback collection will help solve operational problems while strengthening the quality of the interface system [12].

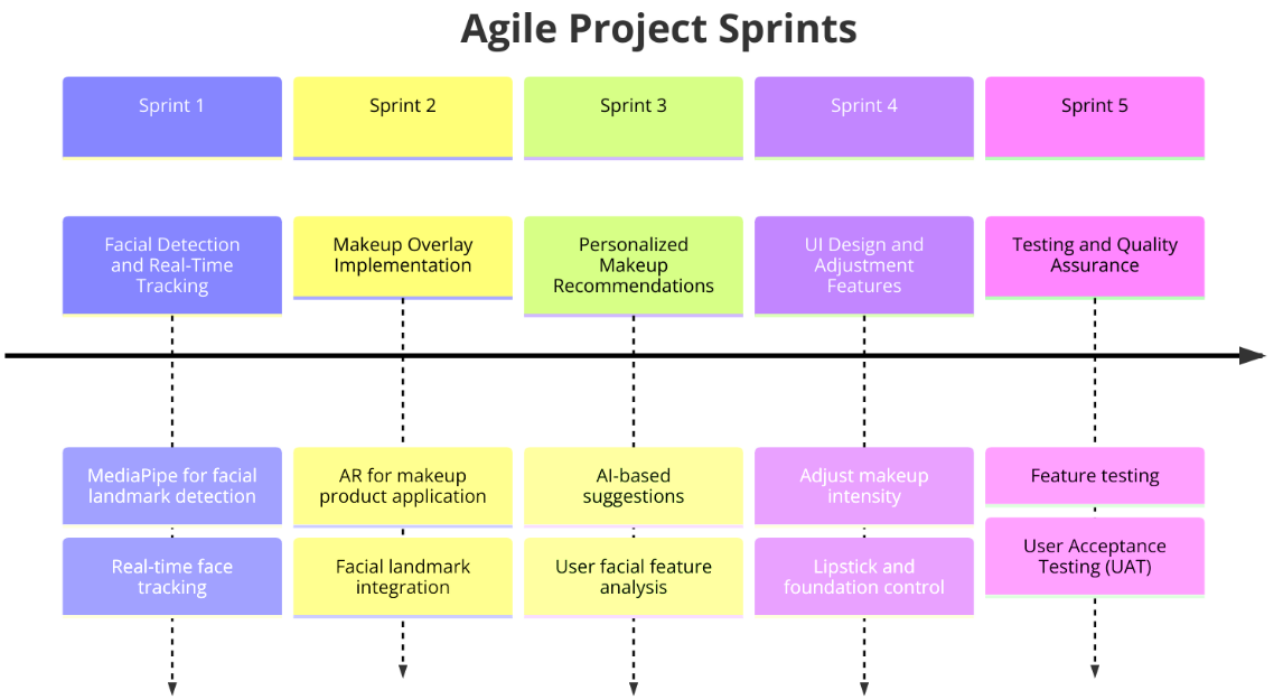


Figure 3.1: Agile Project Sprints

3.3 Testing and Validation

Application tests form an essential phase because they verify correct functioning and user satisfaction. Both system functionality and performance will undergo different testing levels to achieve verification [14, 18].

3.3.1 UNIT TESTING

The focus of unit testing will concentrate on separate discrete modules which include:

- **Facial Detection**: The team will evaluate facial landmark discovery through tests in diverse environmental conditions that include different facial emotions and lighting situations.
- **Makeup Application**: The system will test successful application of makeup according to detected facial landmarks and the user's skin tone.

Edge cases such as dim lighting and rapid head movement were also tested to ensure robustness.

3.3.2 INTEGRATION TESTING

The testing phase checks that different system components operate harmoniously after individual feature testing.:

- The system checks to ensure correct makeup application when facial detection data feeds into its system.
- The testing process confirms that users see suggestions instantly in real-time through correctly integrated UI features.
- Testing validates the uninterrupted execution of data between the user interface frontend component and the AI engine backend and AR makeup system components.

3.3.3 USER ACCEPTANCE TESTING (UAT)

Real users at UAT stage evaluate the application to validate if it satisfies their specified needs and expectations. This phase will focus on:

- **Usability**: Users should find the application easy to navigate as well as the user interface should offer intuitive functionality.
- **Makeup Accuracy**: Checking the accuracy of makeup application, such as the natural blending of foundation, the correct placement of lipstick, and the overall look.

3.3.4 PERFORMANCE TESTING

Performance testing of the application will check its ability to manage:

- **Real-Time Processing**: The system will prove its real-time processing capability by avoiding delays even during simultaneous product loading.
- **Hardware Scalability**: The application was tested on different configurations (8GB vs 16GB RAM, 8th vs 13th Gen i7 processors) to ensure consistent performance across mid- to high-end machines.
- **Load Testing**: The system performance will undergo testing with heavy traffic simulations through load testing to verify its capacity to handle large volumes while maintaining responsiveness.

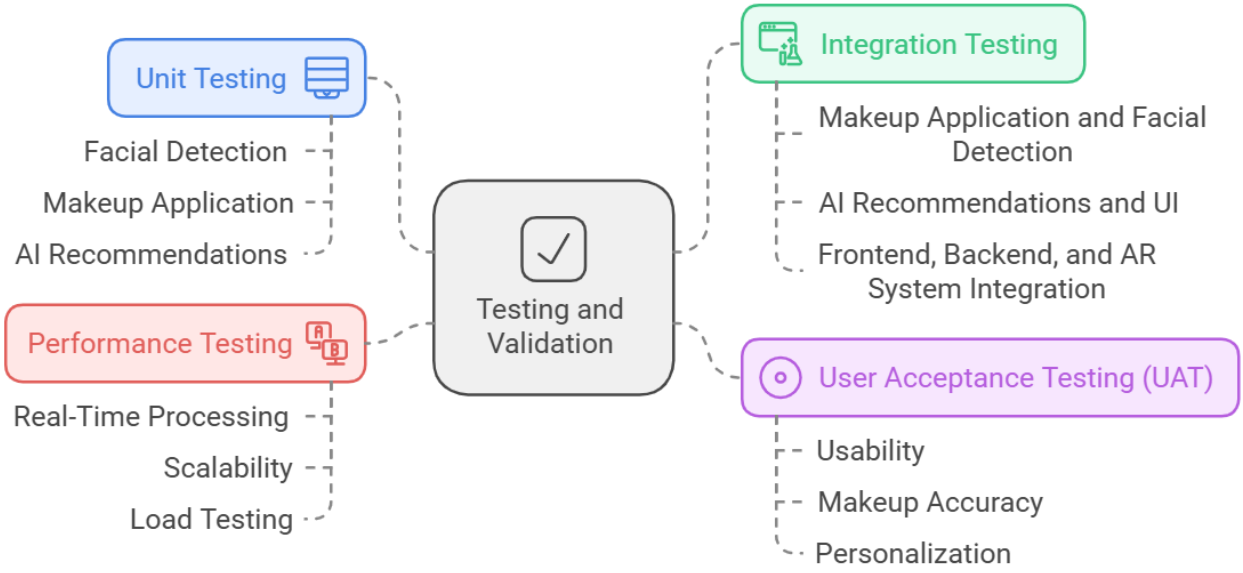


Figure 3.2: Testing And Validation

3.4 CHAPTER SUMMARY

The development of the AI based Virtual Makeup Try-On application is carried out by the use of Agile development which allows development to undergo repetitive cycles with users continually checking the software. It implements the development in a structured method where each feature of the application is divided into separated sprints, which sequentially implement in order to application's features from face detection to AI makeup suggestions and AR make up effects in working, real time on the user.

Since it is made to go through a complete series of unit testing then integration testing then user acceptance testing, the system is function correctly. This system is developed with

agility as the Agile methodology will allow this system to have high adaptability as it will incorporate user feedback all the way to the end of the development cycle. The final product features a combination of AI, AR, and real-time facial detection that allows the users to get an interactive personal virtual makeup application.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 INTRODUCTION

The goal of this chapter is to establish all mandatory system requirements that are necessary for deploying and operating the AI Based Virtual Makeup Try On application. Functional requirements, non functional requirements and hardware requirements which drive to development and operational success need to be segmented into three major categories for the system will be.

- The application demands core behaviors along with essential features which make up its functional requirements.
- The quality attributes of system performance together with security and usability along with scalability form the essential part of non-functional requirements.
- Hardware requirements specify the physical components essential to operate the application by emphasizing both the processing ability for live makeup application and AI algorithms alongside AR visualization capabilities.

4.2 FUNCTIONAL REQUIREMENTS

The functional requirements define exact features along with behaviors which the system needs to support. The system requirements at a basic level define essential operations needed for the AI-Based Virtual Makeup Try-On application to serve its users.

4.2.1 REAL-TIME MAKEUP APPLICATION

Real-time Application of Makeup:

- The system shall use augmented reality (AR) to allow users to apply virtual makeup products in real time.
- The system shall support multiple product categories including lipstick, foundation, eyeshadow, eyeliner, blush, and eyebrow color.

- The system shall overlay makeup effects aligned to the user's facial features, adapting dynamically to facial movements such as smiling, blinking, or head tilting.

Dynamic Adjustments:

- The system shall allow users to adjust the intensity and color of each applied product in real time.
- The system shall support user control over product opacity (e.g., foundation, eyeshadow) and thickness (e.g., eyeliner).

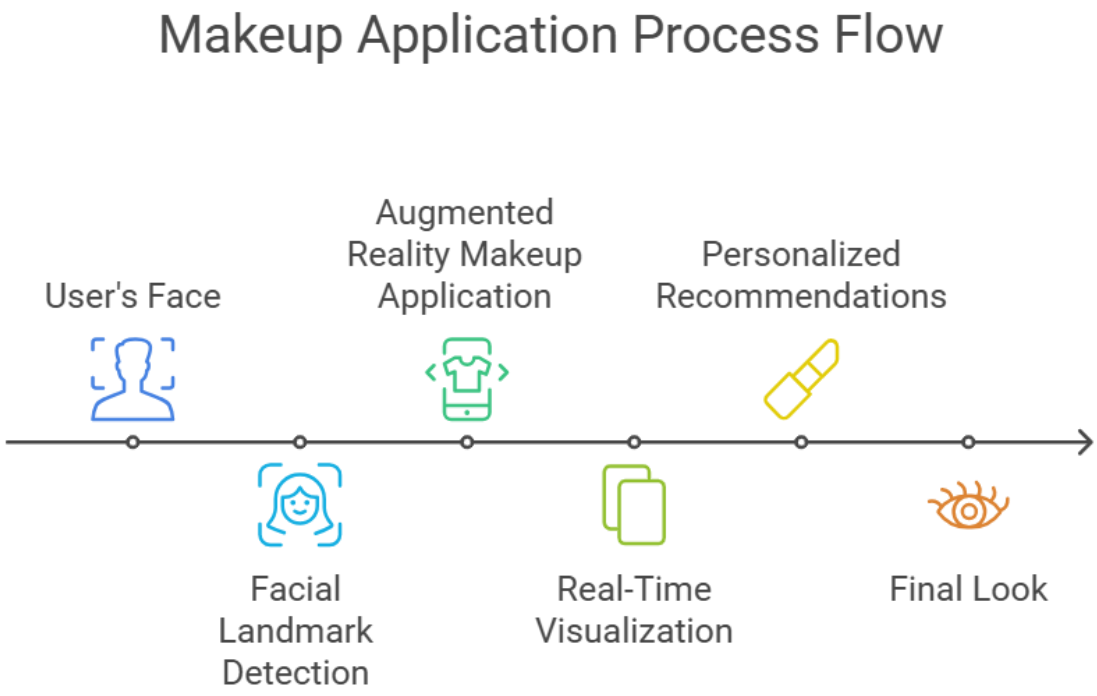


Figure 4.1: Makeup Application Process Flow

4.2.2 Facial Detection and Landmark Tracking

Facial Landmark Detection:

- The system shall implement MediaPipe to detect 468 facial landmarks in real time.
- The system shall identify key facial regions including eyes, lips, eyebrows, nose, cheeks, and chin for accurate makeup placement.

Real-Time Face Tracking:

- The system shall continuously track facial expressions and motions (e.g., smiles, blinks, and rotations) to maintain accurate overlay alignment.

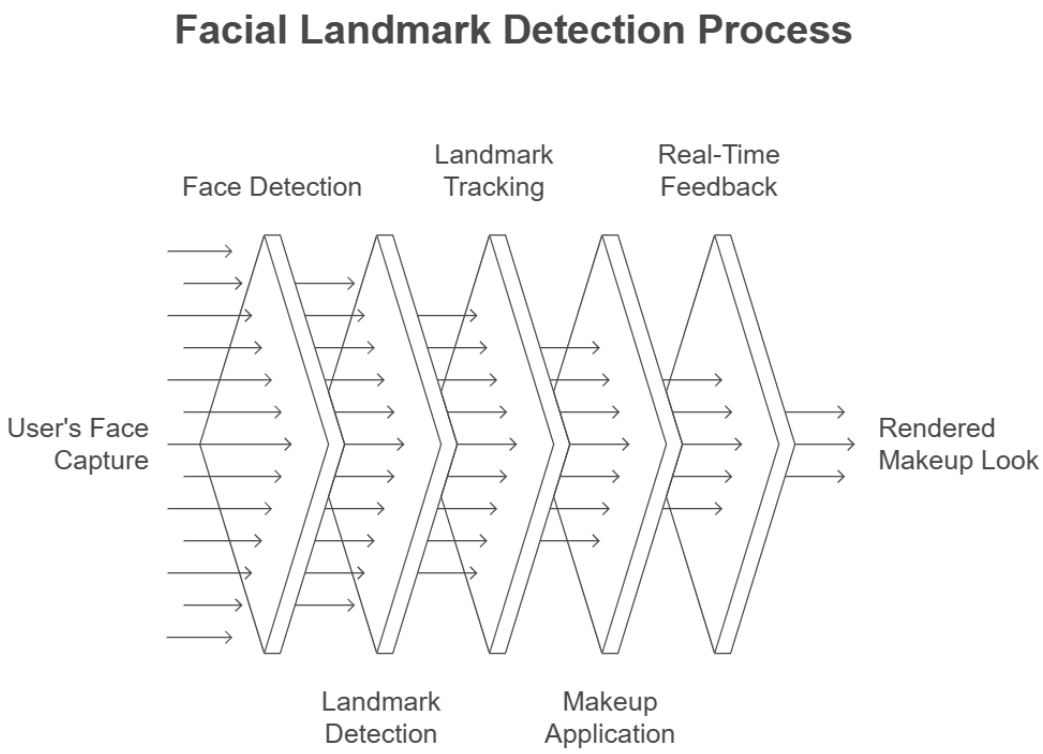


Figure 4.2: Facial Landmark Detection

4.2.3 USER INTERFACE (UI) AND INTERACTION

UI Design:

- The system shall provide a user-friendly interface with clearly labeled product categories and controls.
- The system shall display live camera feed with interactive overlays for selected makeup products.
- The system shall allow users to apply, remove, and switch products using intuitive buttons or sliders.

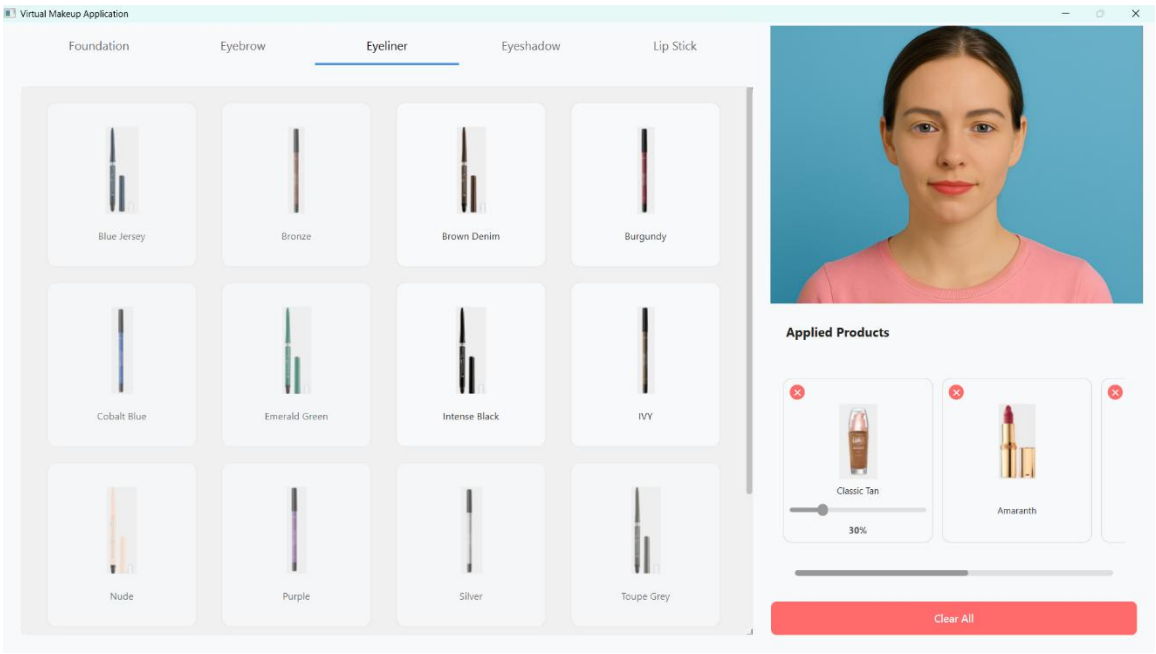


Figure 4.3: Ui Mockup

4.2.4 Multi-Platform Support

Cross-Platform Compatibility:

- The system shall support installation and operation on both Windows and macOS platforms.
- The system shall maintain consistent performance and functionality across supported operating systems.

4.3 Non-Functional Requirements

The non-functional requirements define the quality attributes that the AI-Based Virtual Makeup Try-On application shall meet. These include performance, scalability, usability, and reliability to ensure the system delivers a consistent and effective user experience.

4.3.1 Performance

- The system shall apply virtual makeup overlays with low latency, completing rendering within 500 milliseconds to ensure real-time responsiveness.
- The system shall maintain high throughput by efficiently processing data from the facial recognition system and AR rendering modules.

- The system shall handle continuous facial tracking and product overlays without noticeable delays or frame drops.

4.3.2 SCALABILITY

- The system shall support horizontal scalability, allowing it to accommodate a growing user base and increased processing demands by distributing workloads across multiple processing units or machines.
- The system shall allow for the future integration of additional features, such as new makeup product categories or platform support (e.g., mobile or web).

4.3.3 USABILITY

- The system shall provide a user-centric interface that is accessible to users with varying levels of technical expertise.
- The system shall include support tools such as tutorials, tooltips, or help guides for first-time users.
- The system shall enable users to interact smoothly with the interface, including selecting products, adjusting intensity, and receiving real-time feedback without noticeable lag.

4.3.4 RELIABILITY

- The system shall maintain high availability (99.9%), even during peak usage periods such as online consultations or promotional campaigns.
- The system shall include automatic failover mechanisms to ensure continued operation in the event of infrastructure failure or unexpected system errors.

4.4 SOFTWARE REQUIREMENTS

The specifications provide the requirement of the use of certain tools and frameworks with platforms for building and running the AI Based Virtual Makeup Try On application. This section has specified the necessary software elements required for development and operational need.

4.4.1 OPERATING SYSTEMS

Developers will need different deployment platforms, hence the system has to support a multitude of operating systems [20].

- Windows 10 (or later)
- macOS Mojave (or later)
- Ubuntu 20.04 LTS (or later)

The system is developed by application development team and tested and developed across various operating systems to ensure compatibility and peak performance.

4.4.2 DEVELOPMENT TOOLS

For building the AI Based Virtual Makeup Try-On application, the following development tools and environment will be used [20]:

IDEs:

- Visual Studio Code for JavaScript and React.js development.
- PyCharm for Python-based backend development and AI model integration.

Programming Languages:

- **Python:** For AI processing, including training models for makeup recommendations and real-time facial recognition.
- **JavaScript:** For frontend development using React.js and backend development using Node.js.

Frameworks:

- **TensorFlow/Keras**: For building and training AI models to provide personalized makeup recommendations based on facial features and skin tone.
- **MediaPipe**: For real-time facial recognition and landmark detection.
- **OpenCV**: For real-time image processing, rendering virtual makeup overlays, and webcam integration.
- **PySide6 (Qt for Python):** For building a modern, interactive desktop GUI. [19].

Libraries:

- **OpenCV**: For additional image processing tasks, such as background removal or facial enhancement.

4.5 NETWORK REQUIREMENTS

The network requirements are such that the application is able to cope with the transmission of real time data for makeup application, facial recognition, and product recommendations. The backend also needs to be able to handle large amounts of data and have an optimal user experience, in addition to all that mentioned above.

4.5.1 INTERNET CONNECTIVITY

- **Stable Internet Connection**: Your video streams and live-face computing need at least a 5Mbps Internet connection to work correctly.
- **High-Speed Internet**: Recommended for optimal performance, especially for large-scale deployments with high user volume (e.g., cloud-based services or retail stores).

4.5.2 BANDWIDTH

- **Low Latency**: The program demands quick data movement between the interface and system to deliver live makeup effects and detect faces effectively. The system must handle fast data movements to keep AR graphics up-to-date.
- **High Bandwidth**: Systems need extra data transfer speed because many customers access the system at the same time during retail store and cosmetic sessions.

4.6 SYSTEM ARCHITECTURE AND DEPENDENCIES

The AI-Based Virtual Makeup Try-On application follows a standard client-server system design method. Thefrontend lets users interact whilebackend processes data and runs AI-systems that show live makeup applications.

4.6.1 DEPENDENCIES

- **TensorFlow** and **Keras**: TF and Keras develop AI systems that suggest personalized makeup items from facial characteristics.

- **MediaPipe**: Used for real-time facial recognition and landmark detection during makeup application.

4.7 ENVIRONMENTAL REQUIREMENTS

The environment specifications guarantee that our system runs appropriately when producing and developing software products.

4.7.1 DEVELOPMENT ENVIRONMENT

- **IDEs**: We will build our application in Visual Studio Code and PyCharm for Python developer tools.
- **Programming Environments**: Developers need Node.js runtime and Python 3.x tools installed on their computers.

4.7.2 PRODUCTION ENVIRONMENT

- **Web Hosting**: The application will use cloud infrastructure from AWS or Google Cloud Platform to achieve better availability along with easy expansion capabilities.
- **Browsers**: A browser of Google Chrome, Safari, Mozilla Firefox is required for our application to use to run every internet device platform.

4.8 COMPATIBILITY AND PLATFORM REQUIREMENTS

The system must run on the personal computers and mobile devices to cater to users from various platforms.

4.8.1 CROSS-PLATFORM COMPATIBILITY

- **Desktop**: The application works fine on all specialist operating systems (PC, Macs, and Linux).

4.8.2 BROWSER COMPATIBILITY

- **Browsers**: The application should have good user access on internet platforms by working well with modern web browsers such as Chrome, Firefox Safari.

4.9 CHAPTER SUMMARY

This chapter outlined the key system requirements for the AI-Based Virtual Makeup Try-On application. It covered both functional and non-functional specifications that define how the system shall operate and perform.

The functional requirements focused on real-time makeup application, facial landmark tracking, user interaction, and cross-platform support. Non-functional aspects included performance, scalability, usability, and reliability to ensure a smooth and responsive user experience.

Development tools such as MediaPipe, OpenCV, and PySide6 were selected to meet these goals efficiently. These specifications provide a clear foundation for building a user-friendly, visually accurate, and scalable virtual makeup system.

CHAPTER 5

DESIGN AND ARCHITECTURE

5.1 INTRODUCTION

The design and architecture of the AI Based Virtual Makeup Try On desktop application is discussed in this chapter in detail. The application is built for the purpose of allowing users to virtually apply makeup in real time using Augmented Reality (AR) with facial recognition powered by AI. The system architecture, frontend and backend design, database models, AI integration and security considerations are covered in this chapter to make the application runs smoothly.

As the project will be a desktop application, we are going to use such technologies and tools with a desktop aspect as application for AR and AI.

5.2 SYSTEM ARCHITECTURE

The client (desktop) based architecture of the AI Based Virtual Makeup Try On application does all processing such as facial recognition, makeup application, and recommend using AI locally on the users machine. The assumed offline operation of the system is guaranteed by the architecture, as is the case in many industries with many desktop applications.

5.2.1 HIGH-LEVEL ARCHITECTURE

Desktop Application (Client-Side):

The backend operations (like AI processing, face detection, etc.) are done using Python and the graphical interface (as it involves the GUI) is built with C++/Qt or PyQt [21].

Local within the machine will power media pipe, opencv and tensorflow to power face detection and makeup overlay.

Locally, AI based makeup product recommendations will be generated with pre-trained TForesor/Keras models.

Key Components:

- Frontend: These frameworks have been built on the frontend on Qt or PyQt (for Python based applications) that provides powerful UI tools for the desktop applications like window management, GUI components, and integration with Python libraries.
- Backend: Python uses libraries like OpenCV for image processing as well as TensorFlow or Keras for model inference when backed by AI and the facial recognition in the backend.

5.2.2 DESKTOP APPLICATION FLOW

User Input:

- After the image is uploaded or it becomes available (for live facial detection), the user will create a style sphere.

Facial Detection:

- Facial landmarks are detected and tracked in real time using either the use of MediaPipe or OpenCV.

Makeup Overlay:

- In the application, AR is used to overlay makeup products on the user's face in real time.

User Interaction:

- Users can change the look of makeup intensity, apply more than one product, and save or export the look.

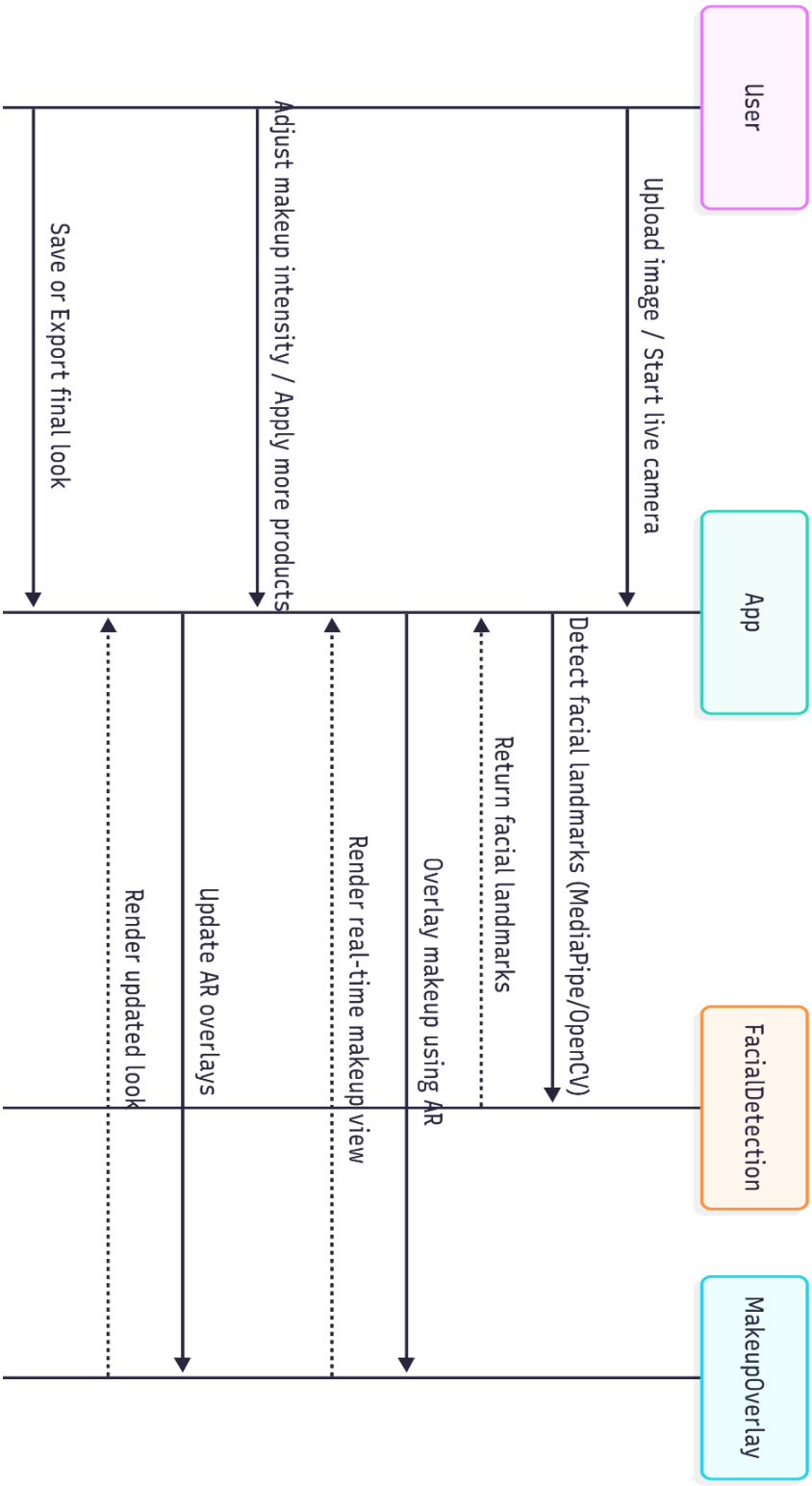


Figure 5.1: Desktop Application Architecture

5.3 FRONTEND DESIGN

The desktop application has an interactive UI, which is a part of the frontend of the desktop application, that enables users to test various makeup products on their facess and manipulate any control.

5.3.1 UI/UX DESIGN

Qt/PyQt Framework:

With the help of PyQt or Qt, we shall create a responsive and user friendly interface which includes buttons, sliders, and components like image viewing.

Users can adjust interactively; the UI will display the user's face with an overlay of makeup products (lipstick, eyeshadow, blush, etc.).

User Controls:

They will be able to drag and drop makeup products, use sliders to change intensity, and real time preview their makeup.

It will have an interface for saving makeup looks, comparing products and testing multiple products at a time.

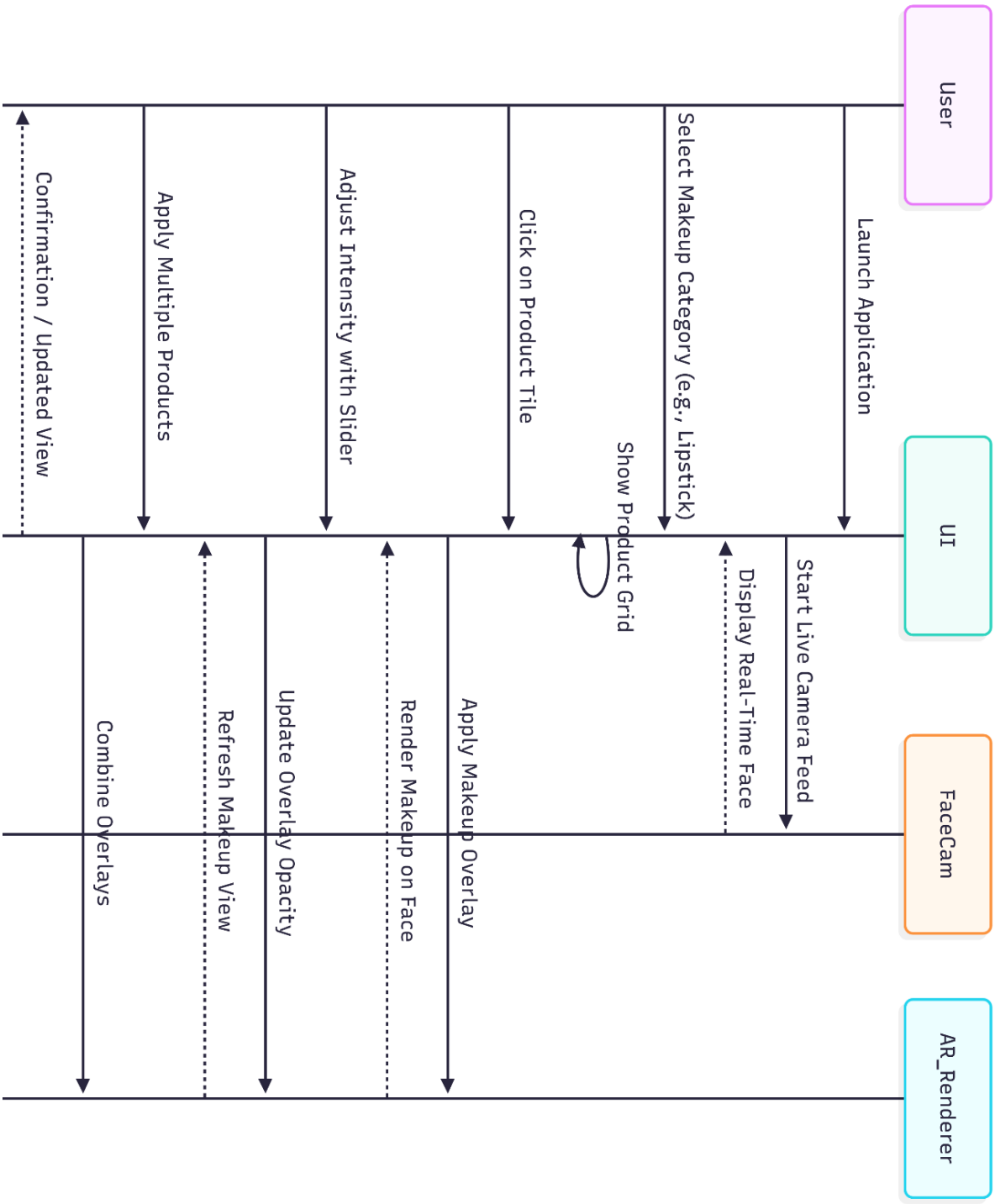


Figure 5.2: Frontend Interaction Diagram

5.4 BACKEND DESIGN

Core AI processing, facial detection, and makeup recommendations, will be handled by the backend of the application.

5.4.1 AI INTEGRATION

Facial Recognition:

For real time makeup application, facial landmarks (eyes, lips, nose) will be detected and tracked by OpenCV or MediaPipe.

The makeup recommendations based on facial features and skin applied from the AI model (powered by TensorFlow/Keras) will be personalized.

Image Processing:

AR will be applied to render the user's makeup products using methods like virtual overlay, lighting correction, skin blending, etc.

5.4.2 BACKEND FLOW

- Facial input (photo or webcam feed) is given by the user.
- The facial features are detected and tracked by the MediaPipe.
- The features are processed by the backend AI model (TensorFlow/Keras) and then it provides the makeup products.
- AR rendering techniques are used to overlay makeup.
- User data saves, preferences saves, and products suggested is handled by the backend.

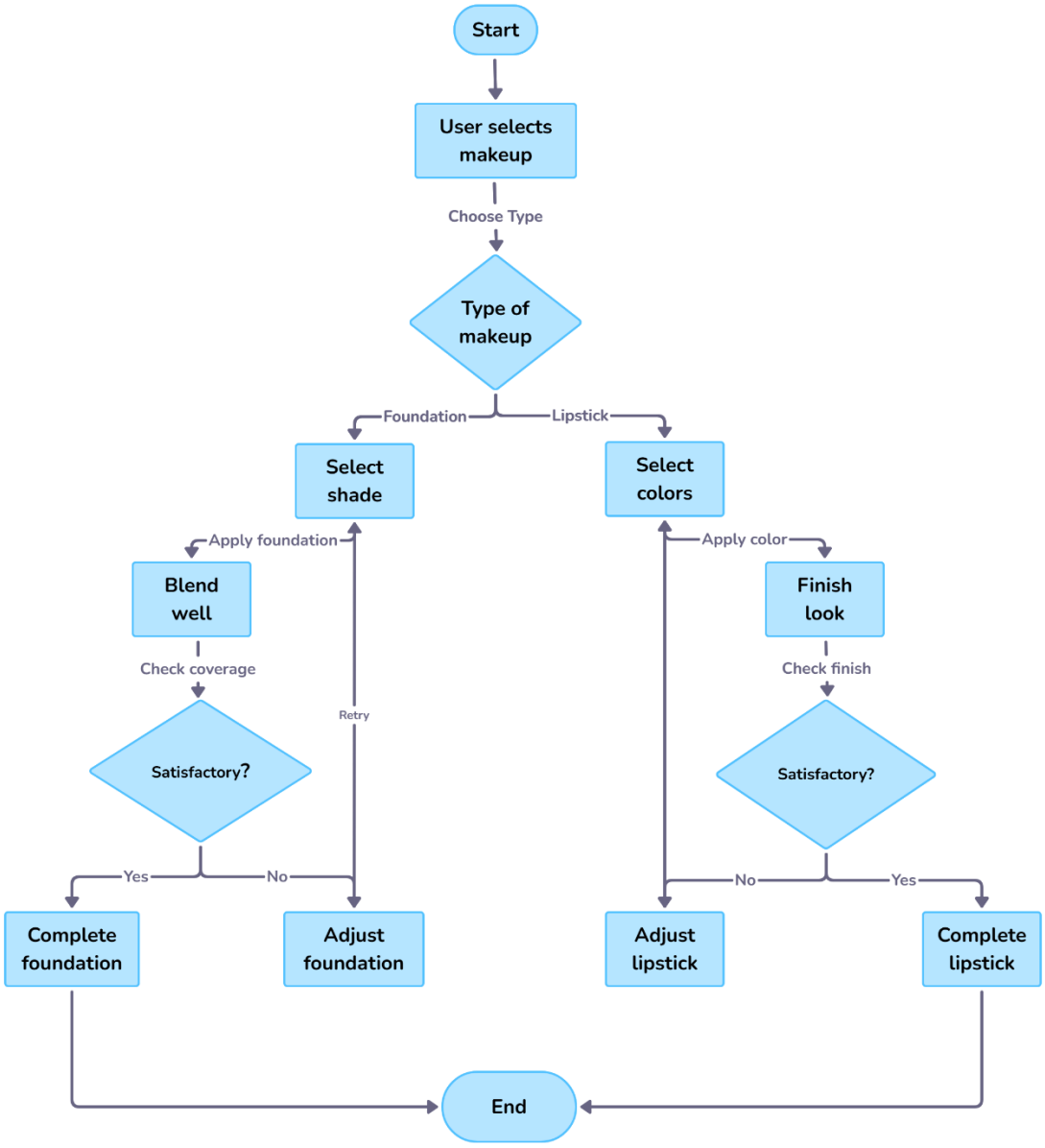


Figure 5.3: Backend Ai Model Flow Diagram

5.4.3 SYSTEM FLOW AND DATA PIPELINE

This system flow diagram illustrates the real-time data pipeline used in the AI-Based Virtual Makeup Try-On desktop application. It shows how the frontend, camera input, facial recognition module, and AR rendering work together to generate the live augmented makeup view.

Workflow Explanation

User Interface – PySide6

The user interacts with a GUI built using PySide6. This frontend allows the user to select makeup categories, adjust intensity, and preview results.

- **Webcam Feed**

A live video feed is captured from the user's webcam, which continuously sends facial frames to the backend.

- **Facial Detection – MediaPipe**

MediaPipe detects 468 facial landmarks from each video frame and generates a landmark map of key facial features (eyes, lips, cheeks, etc.).

- **Coordinate Mapping & Rendering – OpenCV**

The detected landmark coordinates are passed to the Makeup Overlay Renderer, which uses OpenCV to apply selected makeup effects (lipstick, eyeliner, etc.) aligned to the user's facial structure.

- **Camera Display Output**

The final result is rendered in real time, showing the user a live camera preview with AR-based virtual makeup applied.

- **External Libraries Used**

MediaPipe: For facial landmark detection

OpenCV: For drawing overlays and rendering AR effects

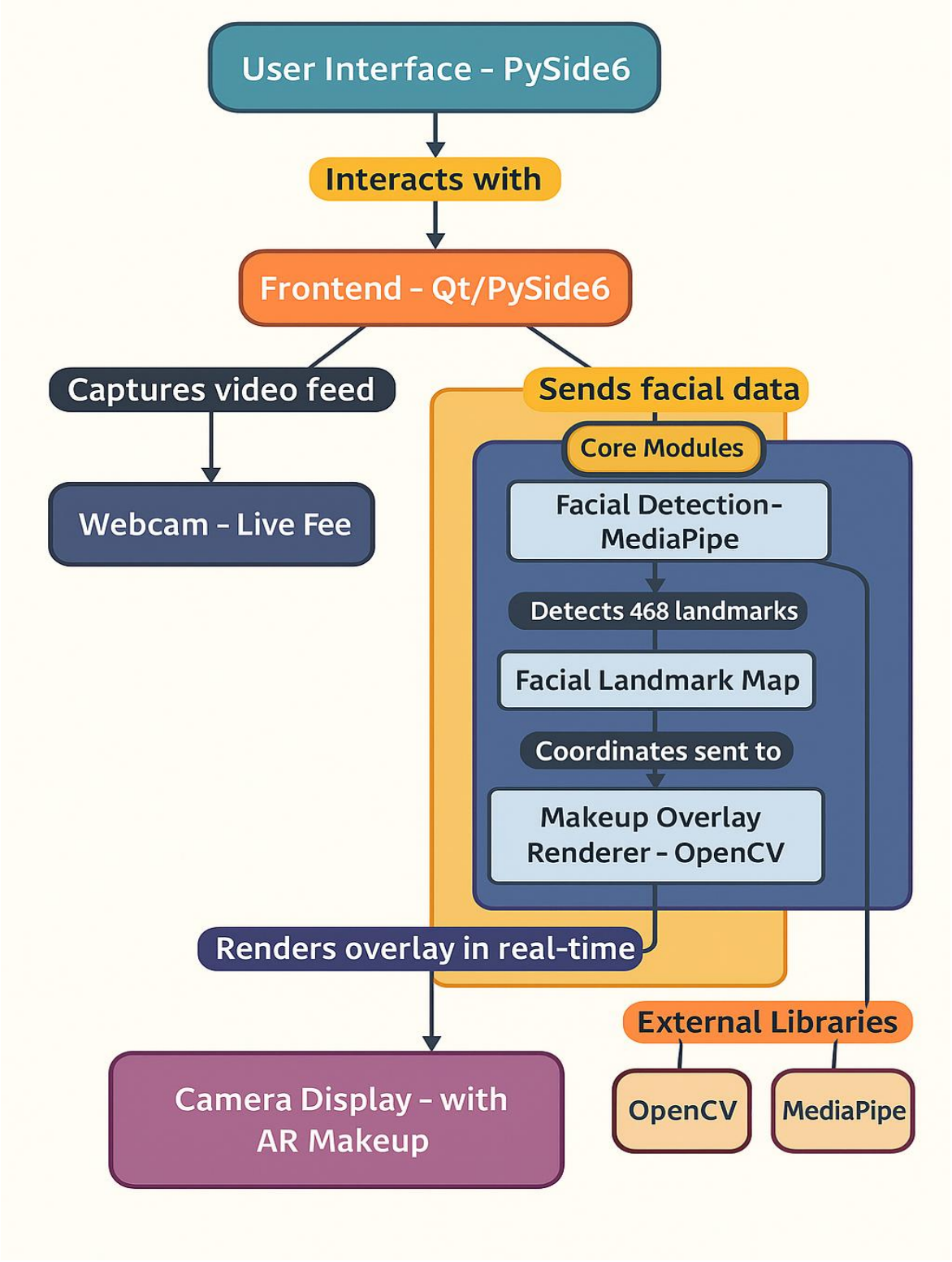


Figure 5.4: System Flow Diagram

5.5 FULL-DRESS USE CASE

To better understand the interaction between the user and the AI-Based Virtual Makeup Try-On system, a use case diagram has been designed. This diagram identifies all the major functionalities that the system provides to its primary actor—the User—and represents how the user interacts with each feature.

The system allows the user to perform actions such as selecting a makeup category (lipstick, eyeliner, foundation, etc.), adjusting product intensity, applying multiple products simultaneously, saving a look, or resetting the session. These use cases collectively define the functional behavior of the application from the user's perspective.

5.5.1 FULL-DRESS USE CASE #1: LAUNCH APPLICATION

Use Case Name	**Launch Application**
Actor	User
Goal	To start the AI Virtual Makeup Try-On application
Preconditions	The application is installed on the user's desktop
Postconditions	The application UI is loaded and ready for interaction

Table 5.1: Full-Dress Use Case #1

Main Success Scenario

1. User double-clicks on the application icon.
2. System loads all required modules and resources.
3. System initializes the user interface (PySide6).
4. Main screen is displayed, and camera access is requested.

- 2a. Required module missing → Show error and exit.
- 4a. Camera access denied → Show warning and disable AR features.

5.5.2 Full-Dress Use Case #2: Start Live Camera Feed

Use Case Name	**Start Live Camera Feed**
Actor	User
Goal	To activate the webcam and begin real-time face tracking
Preconditions	Application is launched and camera is connected
Postconditions	Live video feed is shown in the UI

Table 5.2: Full-Dress Use Case #2

Main Success Scenario

1. User clicks "Start Camera" or system auto-activates camera on launch.
2. System accesses the webcam.
3. System displays the real-time camera feed in the interface.
4. Facial detection module (MediaPipe) starts tracking facial landmarks.

- 2a. Camera not detected → Display "No camera found" message.

5.5.3 Full-Dress Use Case #3: Select Makeup Category

Use Case Name	**Select Makeup Category**
Actor	User
Goal	To choose a specific makeup category for application (e.g., lipstick, eyeliner)
Preconditions	Live camera feed and face tracking are active
Postconditions	Product options related to the selected category are displayed

Table 5.3: Full-Dress Use Case #3

Main Success Scenario

1. User clicks on a tab or menu to choose a category (e.g., Lipstick, Eyeshadow).
2. System displays all available products under the selected category.
3. User proceeds to apply a product.

- 1a. No category selected → System waits for user input.
- 2a. No products in category → Show "No products available" message.

5.5.4 Full-Dress Use Case #4: Apply Makeup Product (Generic – used for all types)

Use Case Name	**Apply Makeup Product**
Actor	User
Goal	To apply a virtual makeup product on the user's face using facial landmarks
Preconditions	Category is selected and face is detected
Postconditions	Selected product is overlaid in real-time

Table 5.4: Full-Dress Use Case #4

Main Success Scenario

1. User selects a product (e.g., red lipstick).
2. System maps the facial landmarks.
3. System renders the selected product on the face in real time using OpenCV.
4. Product is visible in the live preview.

- 1a. Invalid product file → Show fallback default product.
- 2a. Face landmark detection fails → Prompt user to reposition face.

5.5.5 Full-Dress Use Case #5: Remove or Reset Products

Use Case Name	Remove or Reset Products
Actor	User
Goal	To remove applied makeup product(s) or reset to a clean face
Preconditions	At least one product is applied
Postconditions	All or selected product overlays are removed from the preview

Table 5.5: Full-Dress Use Case #5

Main Success Scenario

1. User clicks "Remove" next to an applied product.
2. System clears the corresponding overlay(s).
3. Camera feed updates with a bare face.

5.5.6 Full-Dress Use Case #6: Adjust Product Intensity/Shade

Use Case Name	Adjust Product Intensity/Shade
Actor	User
Goal	To fine-tune the intensity, opacity, or shade of an applied makeup product
Preconditions	At least one product is applied and visible in the live preview
Postconditions	The product appearance is updated based on user adjustments

Table 5.6: Full-Dress Use Case #6

Main Success Scenario

1. User selects an applied product.
2. System displays adjustment controls (slider or shade picker).
3. User adjusts opacity, shade, or intensity.

4. System updates the overlay in real time.
5. Adjusted product is reflected in the live preview.

- 2a. Adjustment tools fail to load → Show error and reload UI.
- 4a. Lag during update → Lower rendering quality temporarily to maintain speed.

5.5.7 Full-Dress Use Case #7: Combine Multiple Products

Use Case Name	**Combine Multiple Products**
Actor	User
Goal	To apply more than one makeup product simultaneously (e.g., lipstick + eyeshadow + eyeliner)
Preconditions	Face is detected and at least one product is already applied
Postconditions	Multiple overlays are rendered on the user's face without conflicts

Table 5.7: Full-Dress Use Case #7

Main Success Scenario

1. User applies a first product (e.g., eyeliner).
2. User navigates to another category (e.g., lipstick).
3. System renders the second product without removing the first.
4. Both (or more) products are displayed together in the live preview.

- 3a. Rendering conflict → System prioritizes product layers or prompts user to adjust.

5.5.8 Full-Dress Use Case #8: View Applied Products

Use Case Name	**View Applied Products**
Actor	User

Goal	To see a summary or list of currently applied makeup products
Preconditions	At least one product has been applied
Postconditions	A panel or list is shown with all active products and their settings

Table 5.8: Full-Dress Use Case #8

Main Success Scenario

1. User opens the “Applied Products” panel.
2. System lists all active makeup layers (e.g., lipstick: red, eyeliner: black).
3. User can view, adjust, or remove any product from the list.

- 2a. Panel does not load → Retry and refresh product state.
- 3a. Product data out of sync → System refreshes from current overlay. |

5.5.9 Full-Dress Use Case #9: Exit Application

Use Case Name	**Exit Application**
Actor	User
Goal	To close the application safely
Preconditions	Application is running
Postconditions	All processes are terminated; application exits cleanly

Table 5.9: Full-Dress Use Case #9

Main Success Scenario

1. User clicks the "Exit" or close button.
2. System prompts to confirm (optional).
3. Application stops camera feed and backend processes.
4. Application window closes.

- 2a. User cancels → Return to active session.
- 3a. Background thread fails to stop → System force-terminates safely.

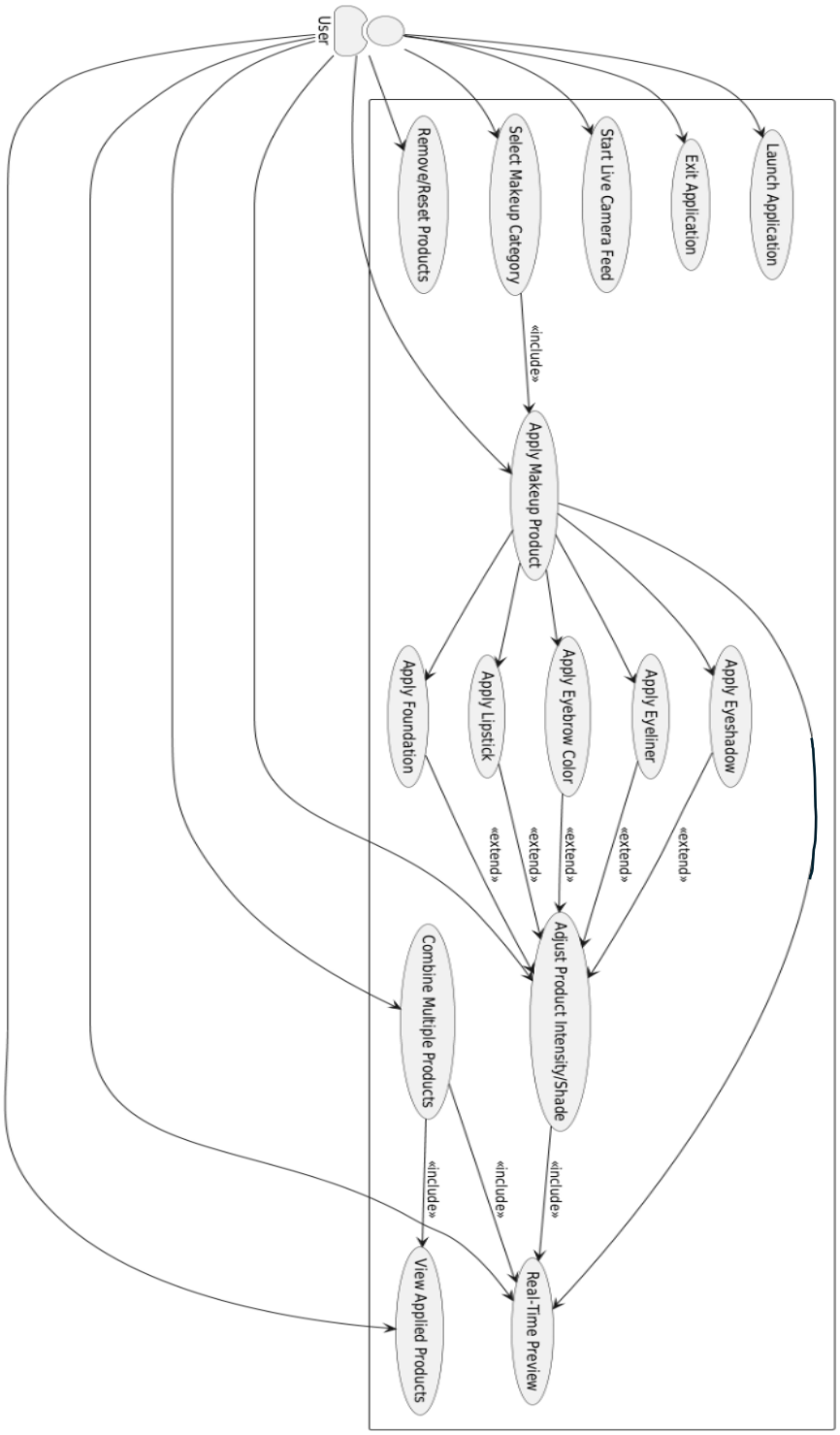


Figure 5.5: Use Case Diagram

5.6 INTEGRATION AND COMMUNICATION

In this section, we explain how all components for the AI Based Virtual Makeup try — on application work and communicate with each other to give a user a perfect experience. Real time makeup application, facial recognition, and product recommendation integration relies on a high degree of integration between frontend, backend, database, and AI models.

5.6.1 FRONTEND AND BACKEND COMMUNICATION

Communication Protocol:

Client (front Qt/PyQt) is talking to the service provider (back Python) through local API calls or functions calls. In the case of real time AR makeup application, the backend processes the facial data and provides the makeup recommendations to the frontend which is later rendered on the screen.

5.6.2 AI MODEL INTEGRATION

Personalized makeup product recommendations are generated from the backend and the AI (TensorFlow/Keras) model is integrated into the backend.

The system understands when the user uploads an image or uses the camera for live real time makeup trials and detects the facial features then transmits it to the AI model.

It then detects the features and generates recommendations based on them and return the recommendations to the frontend for overlaying on the user's face.

5.6.3 REAL-TIME MAKEUP OVERLAY

- With AR techniques, the system showcases makeup products on the user's face in real time. OpenGL or WebGL is used for rendering of the makeup and it is integrated with the Qt interface for desktop apps.
- MediaPipe detects the landmarks from the face and adjusts the intensity and position of the makeup products accordingly on the frontend.

5.6.4 Integration Flow Diagram

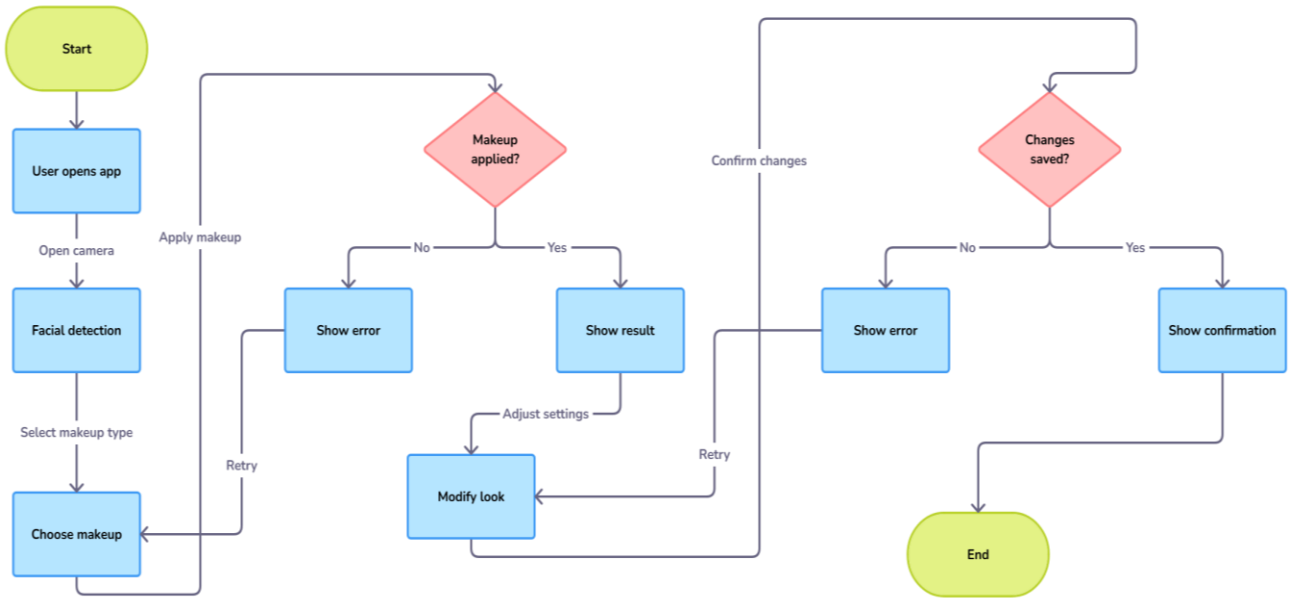


Figure 5.6: System Integration Flow Diagram

5.7 Scalability and Future Enhancements

With application growing, scaling out starts to become necessary to support larger datasets, more makeup products, and more features. It then goes on to discuss how the system is made to scale and potential future improvements.

5.7.1 Scalability Considerations

Desktop Application Scalability:

The data storage (SQLite) and AI models are optimized to manage a bigger databases (more makeup products, more AI models). But desktop applications are normally not as scalable as cloud based systems.

In local scaling, think about how to optimize AI Model loading and Data Retreival in terms of local storage optimisation (cache makeup product data, reduce real time Inference times).

5.7.2 Handling Larger User Base

For each increasing number of users, the application should be able to handle such an increase.

If the desktop app moves towards supporting a large scale enterprise deployments, database management can be scaled by sharding or with the use of distributed databases.

If cloud storage and cloud AI processing is needed in the future product versions then it can also be used.

5.7.3 FUTURE ENHANCEMENTS

To enable future enhancements, the system's flexibility is built into it.

More Makeup Products: Adding new product categories (e.g., eye lenses, nail polish, etc.) to the makeup catalog.

Mobile App Integration: Develop an iOS and Android Mobile Application that integrates some real time AR features from desktop version.

Extended AI Capabilities:

AI can further be made more accurate by skin tone matching or personalized makeup tutorials and even AI generated makeup designs (for example, the use of Generative Adversarial Networks (GANs) to generate an entirely new makeup look that the human created).

5.7.4 Scalability Flow Diagram

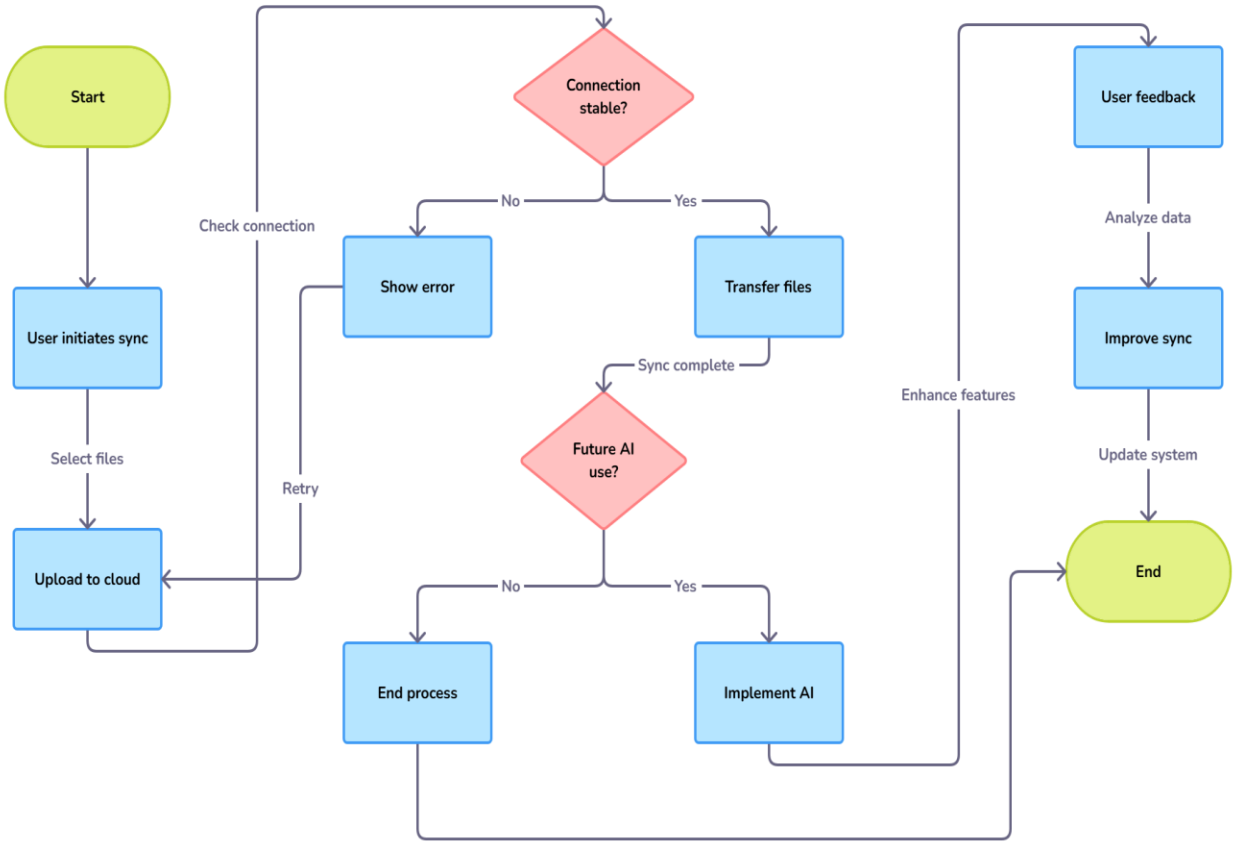


Figure 5.7: Scalability And Future Enhancement Diagram

5.8 Chapter Summary

In this chapter we have described the architecture and the design of the AI Based Virtual Makeup Try On desktop application. The system offers a seamless, secure, and interactive experience to users utilizing a client server model, the integration of AI recommendations, real time AR Makeup overlays, and local storage solution. The design is visible scalable for future enhancements including cloud integration, mobile apps, as well as other AI powered features.

CHAPTER 6

SYSTEM TESTING

6.1 INTRODUCTION

System testing plays a crucial role in evaluating the overall functionality, stability, and reliability of the AI-Based Virtual Makeup Try-On desktop application. It ensures that the application meets the intended requirements as specified in the design and fulfills the real-world expectations of end users, especially in terms of real-time makeup rendering accuracy and system responsiveness. As the project aims to provide real-time application of virtual makeup such as lipstick, eyeliner, eyeshadow, foundation, and blush using Artificial Intelligence (AI) and Augmented Reality (AR), the testing phase focuses on assessing these core functionalities under different hardware and usage conditions.

The testing phase adopted a manual testing approach, allowing for real-time interaction with the system under practical scenarios [24]. The application was thoroughly tested on Windows operating systems across two hardware configurations:

- **Development Machine**: 8 GB RAM, Intel Core i7 (13th Gen)
- **Test Machine**: 16 GB RAM, Intel Core i7 (8th Gen)

This environment provided insights into system performance on both high- and mid-tier hardware, helping identify potential limitations in responsiveness and rendering consistency.

6.2 TESTING OBJECTIVES

The primary goals of system testing were as follows:

- Validate core functionalities such as real-time facial landmark detection and the accurate overlay of virtual makeup products.
- Assess usability and user satisfaction, particularly ease of navigation and interaction with the GUI.

- Evaluate system performance, ensuring low latency and smooth operation during live facial tracking.
- Detect and document bugs or misalignments in makeup rendering for future refinement.
- Incorporate user feedback, particularly from classmates who acted as external testers, to enhance the overlay quality and visual appeal.

6.3 TYPES OF TESTING PERFORMED

The following types of testing were conducted manually [24]:

6.3.1 FUNCTIONAL TESTING

This involved verifying that all core features of the application worked as intended. Specifically:

- Facial landmarks were correctly detected using MediaPipe in real time [22].
- Virtual makeup products (lipstick, eyeliner, eyeshadow, foundation, blush) rendered correctly and responded to facial movements such as head tilts or smiles.
- Users could adjust the intensity and shade of makeup using UI sliders and options.

Each function was tested individually and in combination to ensure proper layering and application behavior.

6.3.2 USABILITY TESTING

Classmates and non-technical users were asked to interact with the system. The objectives here were:

- To assess the intuitiveness of the UI/UX design.
- To evaluate how easily a new user could select and apply different makeup products.
- To determine whether makeup adjustments (opacity, shade selection, switching between products) were understandable and fluid.

Feedback was collected informally and led to small improvements in the UI and makeup intensity transitions.

6.3.3 PERFORMANCE TESTING

Given the real-time nature of the system, performance testing focused on:

- Frame rate stability during facial tracking and makeup rendering.
- Processing speed when switching products or adjusting intensity.
- Lag or delay in AR overlay when the user moved their face rapidly.

The system generally performed well across both testing machines, with only minor optimizations required in overlay refinement [22].

6.4 DETAILED TEST CASES

To ensure thorough validation of the core functionalities of the AI Virtual Makeup Try-On application, structured manual test cases were created and executed for each individual feature. These test cases were designed to assess the correctness, consistency, responsiveness, and visual accuracy of makeup overlays on various facial movements and lighting conditions.

Each test case includes:

- Test ID
- Description
- Input or Action Performed
- Expected Output
- Actual Output
- Status (Pass/Fail)
- Remarks or Observations

6.4.1 LIPSTICK OVERLAY TEST CASES

Test ID	Test Description	Input/Action	Expected Output	Actual Output	Status	Remarks
TC-LS-01	Apply default lipstick	Select default lipstick from product list	Lipstick applied smoothly on both lips	Lipstick aligned correctly with minor upper lip offset	Pass	Slight blurring on lower lip in dim light
TC-LS-02	Adjust lipstick opacity	Move opacity slider to 50%	Lipstick color becomes semi-transparent	Color intensity reduced as expected	Pass	Works well even during facial expressions
TC-LS-03	Change lipstick shade	Select “Bright Red” shade	Lipstick changes to bright red instantly	Shade reflected accurately	Pass	No delay observed in update
TC-LS-04	Smile with lipstick applied	Smile while wearing lipstick	Lipstick remains stable and doesn’t misalign	Lipstick shifts slightly on corner of mouth	Pass	Improvement recommended for smile detection adaptation

Table 6.1: Functional Test Cases For Lipstick Overlay

6.4.2 EYELINER OVERLAY TEST CASES

Test ID	Test Description	Input/Action	Expected Output	Actual Output	Status	Remarks
TC-EL-01	Apply basic eyeliner	Select “Basic Black” eyeliner style	Eyeliner appears on upper eyelid in black color	Eyeliner applied correctly	Pass	Natural blending effect shown
TC-EL-02	Change eyeliner thickness	Increase thickness to 75%	Thicker line applied on eyelid	Thickness updated accurately	Pass	Accurate even during rapid eye movement
TC-EL-03	Blink with eyeliner applied	Blink several times	Eyeliner remains stable and does not flicker	Flickering observed on left eyelid	Pass	Needs smoothing frame logic during blinking

Table 6.2: Functional Test Cases For Eyeliner Overlay

6.4.3 EYESHADOW OVERLAY TEST CASES

Test ID	Test Description	Input/Action	Expected Output	Actual Output	Status	Remarks
TC-ES-01	Apply default eyeshadow	Select “Neutral Brown” from palette	Eyeshadow blended over upper eyelid	Displayed as expected	Pass	Slight edge overlap on small eye sizes
TC-ES-02	Change eyeshadow intensity	Set to 30% intensity	Light eyeshadow tint visible	Tint appeared as expected	Pass	Good gradient control

TC-ES-03	Head movement with eyeshadow applied	Turn head left and right	Eyeshadow position tracks accurately	No misalignment during movement	Pass	Excellent landmark stability

Table 6.3: Functional Test Cases For Eyeshadow Overlay

6.4.4 FOUNDATION (BASE) OVERLAY TEST CASES

Test ID	Test Description	Input/Action	Expected Output	Actual Output	Status	Remarks
TC-FD-01	Apply light foundation shade	Select "Light Beige" tone	Foundation blended across full face with even tone	Even application except under jaw area	Pass	Improvement suggested for jawline blending
TC-FD-02	Adjust coverage to full	Set opacity to 100%	Complete coverage of base with smooth effect	Strong application with natural look	Pass	Realistic rendering with no lag
TC-FD-03	Smile with foundation applied	Smile and move face	Foundation remains consistent during movement	Light flicker during fast movement	Pass	May optimize AR performance during transitions

Table 6.4: Functional Test Cases For Foundation (Base) Overlay

6.5 BUG TRACKING AND RESOLUTIONS

During the testing phase, several minor bugs and inconsistencies were observed in the application's makeup rendering system, primarily related to alignment accuracy,

responsiveness to facial expressions, and occasional latency. Although no critical system failures were reported, these functional imperfections were documented and addressed systematically to enhance the overall performance and user experience of the AI-Based Virtual Makeup Try-On system.

6.5.1 BUG REPORTING METHODOLOGY

All issues were recorded in a structured manual bug log maintained by the development team. For each issue, the following attributes were noted:

- Bug ID
- Description
- Date of Detection
- Module Affected
- Environment (Test Machine/ Development machine)
- Severity (Low, Medium, High)
- Steps to Reproduce
- Fix Status (Open/Closed)
- Date Resolved
- Remarks

6.5.2 SAMPLE BUG LOG

Bug ID	Module	Description	Severity	Environment	Fix Status	Remarks
BUG-001	Lipstick Overlay	Lipstick shifts slightly from upper lip during smile	Medium	i7 8th Gen, 16GB RAM	Fixed	Facial movement smoothing algorithm improved

BUG-002	Eyeliner	Eyeliner flickers during rapid blinking	Low	i7 13th Gen, 8GB RAM	Fixed	Added frame-level consistency check using buffer smoothing
BUG-003	Foundation	Jawline coverage missing on specific head angles	Medium	i7 8th Gen, 16GB RAM	Fixed	Adjusted jaw landmark mapping and improved boundary blending
BUG-004	Eyeshadow	Slight overlap beyond eyelid boundary on small facial frames	Low	Both Machines	Open	Requires facial shape-specific calibration (future sprint)

Table 6.5: Bug Log Detailing Reported Issues, Severity, And Resolutions

Every fixed bug helped to make visual accuracy and user satisfaction become significantly better. Trained items that were left open were recorded to be refined later during other phases when they need to be re-trained on a machine learning model or when the change in facial geometry would need reconfiguration based on facial diversity.

6.6 User Feedback Summary and Incorporation

The usability and visual fidelity of the application were tested by performing an informal user testing with few classmates and peers. The testers were provided with a short overview of the application and requested to do some common activities to choose a product of makeup, to vary its intensity, to try a combination of various products.

6.6.1 User Testing Process

Number of participants: 7 (3 male, 4 female)

User profile: All were students, none from a technical or cosmetic background

Duration: Each session lasted approximately 10–15 minutes

Devices used: Both test machines (13th Gen i7 – 8GB, 8th Gen i7 – 16GB)

Tasks performed:

- Apply each type of makeup (lipstick, eyeliner, eyeshadow, foundation, blush)
- Modify intensity and color
- Use facial expressions (smile, turn head) and observe overlay behavior
- Give open-ended feedback

6.6.2 FEEDBACK SUMMARY

Feedback Category	User Comments	Action Taken
Overlay Alignment	Lipstick and foundation sometimes misaligned during smile or side head turn	Fixed through improved landmark smoothing
Ease of Use	Users found the UI intuitive and easy to navigate [23]	No major changes required
Realism of Makeup	Makeup looked realistic, especially blush and foundation	Positive — retained current rendering approach
Intensity Control	Eyeliner and eyeshadow opacity adjustments were appreciated	Enhanced slider range for finer control
Performance	Minor lag observed on older device when applying multiple products simultaneously	Backend rendering optimizations applied to reduce latency

Suggestions	Add undo/redo feature for makeup application changes	Marked as future enhancement

Table 6.6: Summary Of User Feedback And Implemented Enhancements

In general, the users felt satisfied with the capabilities of the simulation of applying makeup in a realistic and interactive way in the app. They also made valid contributions to the responsiveness of the makeup overlays and polishing the user interface to interact more easily [23].

6.7 TESTING SUMMARY AND EVALUATION

After the full cycle of system testing in the established conditions, the AI Virtual Makeup Try-On application was found to be quite stable in terms of functionality and visually convincing, highly usable in the conditions in which they were put under the test. The most important evaluation findings are as follows:

Strengths Identified:

- **Accurate Facial Landmark Tracking**: The face mesh model used in MediaPipe was highly accurate and detected the facial landmarks at many different angles and under different lighting settings.
- **Real-Time Responsiveness**: Users experienced minimal delay (< 0.5 seconds) in rendering overlays even with continuous face movement.
- **Stable Multi-Product Application**: Users could apply multiple products simultaneously (e.g., lipstick + eyeshadow + blush) without performance degradation.
- **User Engagement**: Testers spent time experimenting with different combinations, indicating strong engagement and interest in the application's features.

Areas for Improvement:

- **Corner-case Blending Issues**: Some users experienced minor glitches around facial edges (e.g., jawline for foundation, outer eyes for eyeshadow).
- **Facial Movement Adaptation**: Minor misalignment still occurred during rapid head movements or exaggerated facial expressions.

- **Extended Facial Diversity**: Testing across broader facial types (age, ethnicity, face shapes) is needed for generalized AI model accuracy.

6.8 CHAPTER SUMMARY

The current chapter gave an extensive explanation to the system testing activity performed on the AI-Based Virtual Makeup Try-On application. The project team conducted a systematic hand-on testing and confirmed the functionality, usability and responsiveness of the system on two hardware set ups, thus when put to common usage scenarios the system worked and responded satisfactorily.

The chapter started by stating the purpose and the kind of testing that was done including functional testing, usability testing and performance testing. Individual features such as lipstick, eyeliner, eyeshadow, foundation and blush were unit tested by providing specific test cases where expected vs. actual results, notations, and alignment concerns were noted. Although most of the test cases passed successfully, certain edge conditions showed a slight misalignment or disparity especially in the movement of the face or the differences in the lighting conditions.

The Bug tracking and resolution formed a major part of this stage. A bug log was kept with list of every bug and whether it has been fixed or not. Some minor yet notable changes were made which include smoothing directions of head turns, addressing the issue with eyeliner flickering on blink, and making jawlines look nice with the foundation, among others. More complex geometry bugs or those needed AI model retraining were set to later development cycles.

This further confirmed the usefulness of the application based on user feedback. The subjects liked the overall relevance of makeup overlays and the ease of use and interface, but they also recommended enhancements such as more effective edge blending and undo/redo. The fact that they engaged with the product showed that the product was not only practical technically but also satisfied important expectations regarding how it looked and how easy it was to use.

Finally, the system testing procedure helped a great deal in strengthening the application as well as its quality. Through small adjustments and incorporation of user-focused

improvements, the app went beyond a proof-of-concept to a stable and coherent desktop tool to virtually apply makeup. Lessons learnt and feedback will be used in subsequent iterations, especially with respect to increasing the robustness of face recognition, enhancing overlay methods, and perhaps inclusion of product suggestions engines or otherwise platform enhance spanning the web or mobile.

CHAPTER 7

APPLICATION PROTOTYPE AND USER MANUAL

7.1 APPLICATION OVERVIEW

The AI-Based Virtual Makeup Try-On app is a desktop application, created through an application of Python, OpenCV, MediaPipe, and PySide6. The application enables users to put on various virtual makeup products to their face in real life such as:

- Foundation
- Eyebrow Color
- Eyeliner
- Eyeshadow
- Lipstick

Computer vision algorithms based on the detection of facial landmarks and taken by live video stream of a user directly implement these features [15]. The design of the interface is accomplished through PySide6 that offers a modern interface GUI with the supersmooth quality and reactivity. Use of the app is also locally based on a local machine, so the app does not need an internet connection once installed, which makes it safe and anonymous.

7.2 INSTALLATION REQUIREMENTS

The application was coded and tested in Python 3.11 environment in Anaconda in order to produce a smooth performance. The libraries needed would be handled by either conda or pip as stated in requirements.txt and environment.yml.

7.2.1 HARDWARE REQUIREMENTS

Component	Minimum Requirement
Processor	Intel Core i7 (8th Gen or higher recommended)

RAM	8 GB (16 GB recommended)
Storage	2 GB free disk space
Operating System	Windows 10 / 11 (64-bit)
Webcam	Integrated or external (HD preferred)

Table 7.1: Hardware Requiremnets

7.2.2 SOFTWARE REQUIREMENTS

Python 3.11

Anaconda (for virtual environment management)

Visual C++ Build Tools (for some packages)

7.2.3 REQUIRED PYTHON LIBRARIES

Key libraries used (full list in requirements.txt)requirements:

- mediapipe for facial landmark detection
- onnxruntime-gpu for efficient inference
- opencv-contrib-python for image processing
- PySide6 for GUI
- numpy, matplotlib, scipy for various calculations and rendering
- sounddevice (optional future integration for voice commands)

7.2.4 INSTALLATION STEPS

Install Anaconda

- Download and install Anaconda for Windows.

Create Environment from YAML

- conda env create -f environment.yml

Activate the Environment

- conda activate virtual-makeup
- (Alternative): Use pip to install from requirements.txt:
- pip install -r requirements.txt

Run the Application

- Navigate to the project directory and run:
- python main.py

7.3 LAUNCHING THE APPLICATION

Once the environment has been successfully set up using either Anaconda or pip, launching the application is a straightforward process.

7.3.1 LAUNCH STEPS

Open Anaconda Prompt or Terminal.

Activate the environment:

- conda activate virtual-makeup

Navigate to the project directory where main.py is located.

Run the main Python script:

- python main.py

The application window will open displaying a live camera feed and a modern graphical interface that includes navigation tabs, makeup product tiles, and real-time product preview functionality.

7.4 USER INTERFACE WALKTHROUGH

The application interface is designed with a focus on simplicity and usability. It consists of two main sections:

- **Left Panel**: Product selection and navigation
- **Right Panel**: Live camera feed and applied products

7.4.1 MAIN NAVIGATION TABS

Located at the top of the window, the following tabs allow users to switch between makeup product categories:

- Foundation
- Eyebrow
- Eyeliner
- Eyeshadow
- Lip Stick

These tabs are interactive and update the selection grid below them.

7.4.2 Product Selection Grid

Each tab displays a grid of available makeup products. Clicking on a product applies it in real time to the user's face in the camera feed.

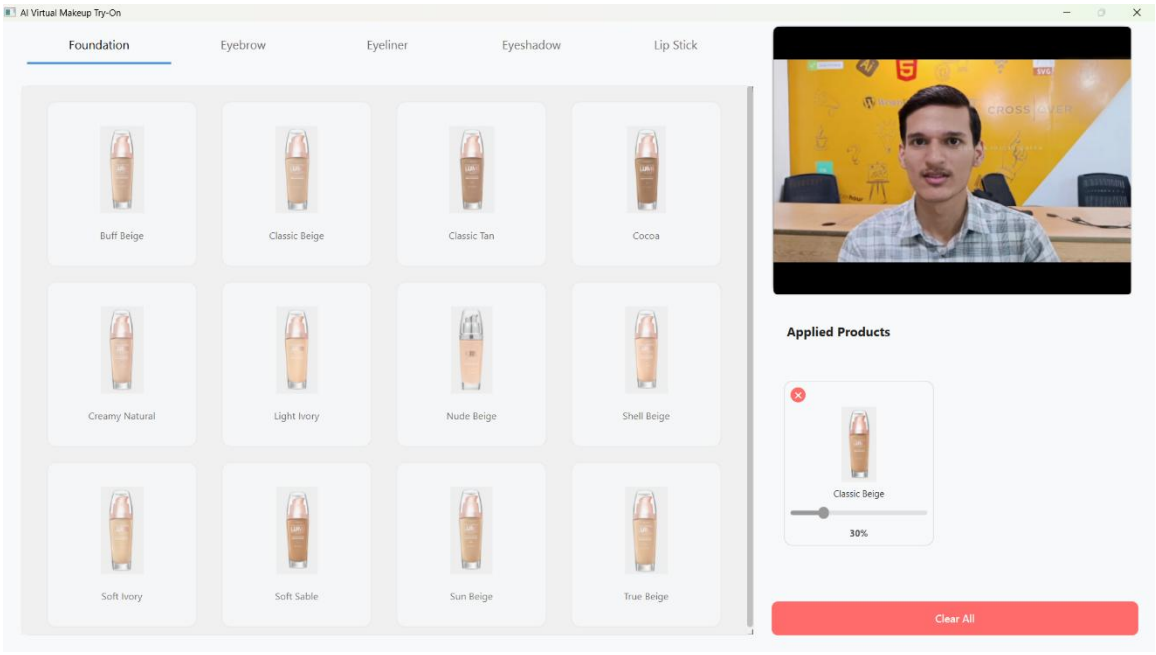


Figure 7.1: Foundation Product Selection Interface

The user selects a shade (e.g., Classic Tan), and the applied product appears on the right with an intensity slider.

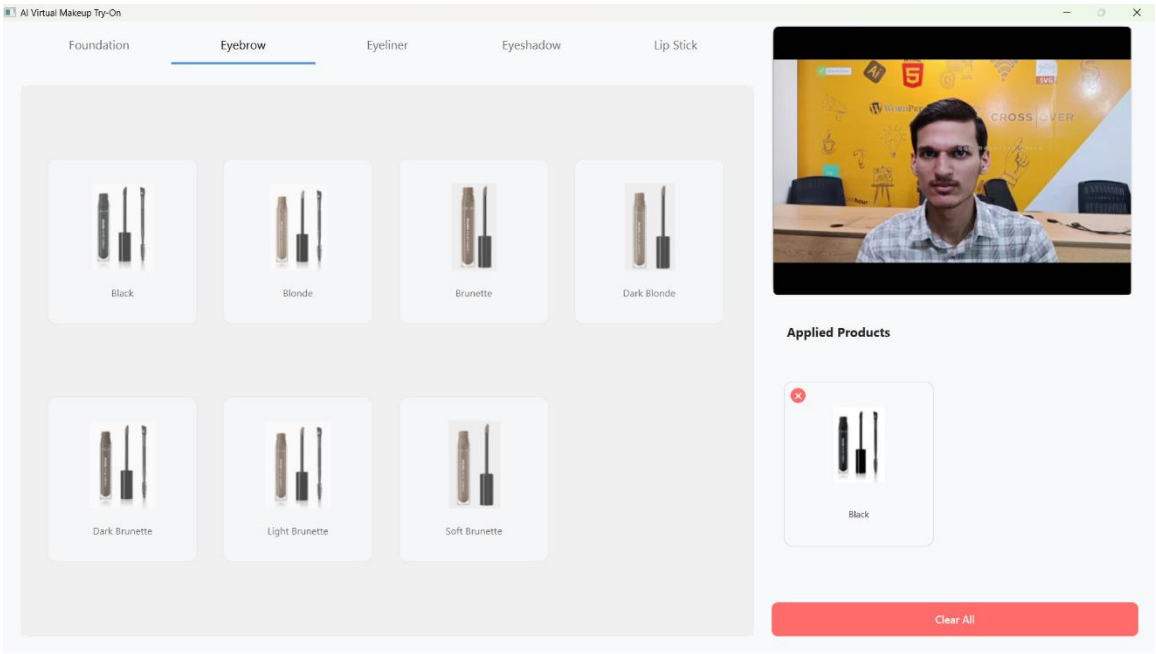


Figure 7.2: Eyebrow Color Selection Interface

Eyebrow colors (e.g., Dark Brunette) are displayed in square cards. Users can apply and remove them individually.

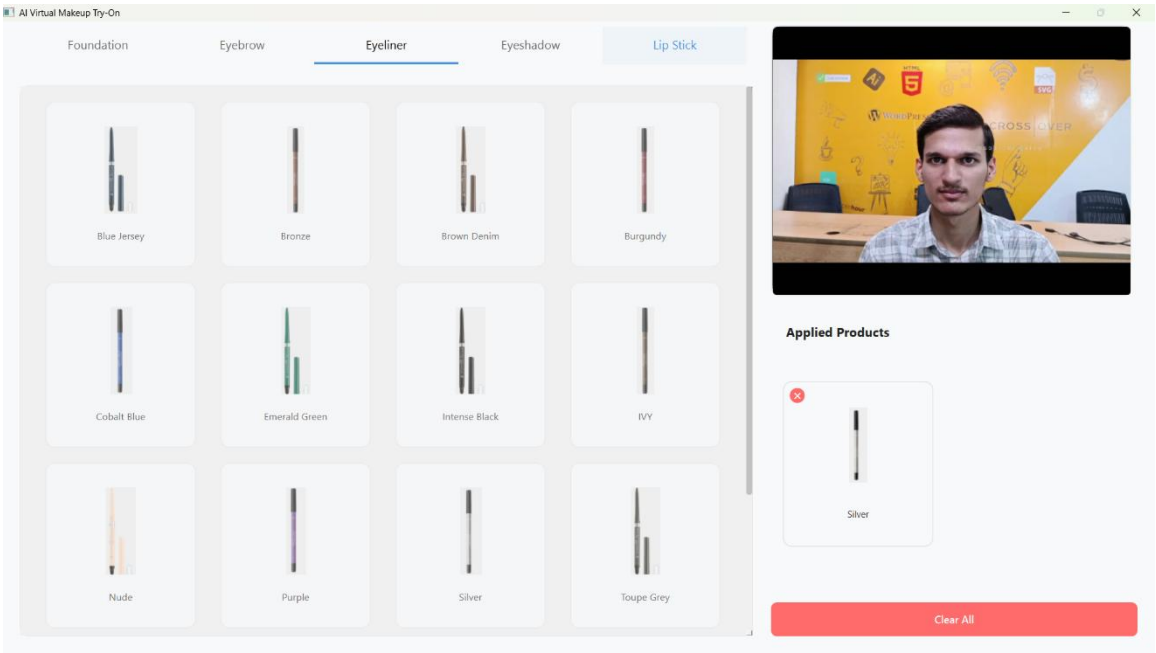


Figure 7.3: Eyeliner Product Interface With Live Preview

A wide range of eyeliners is available, such as “Brown Denim” or “Intense Black”. Eyeliner follows the upper eyelid landmark precisely.

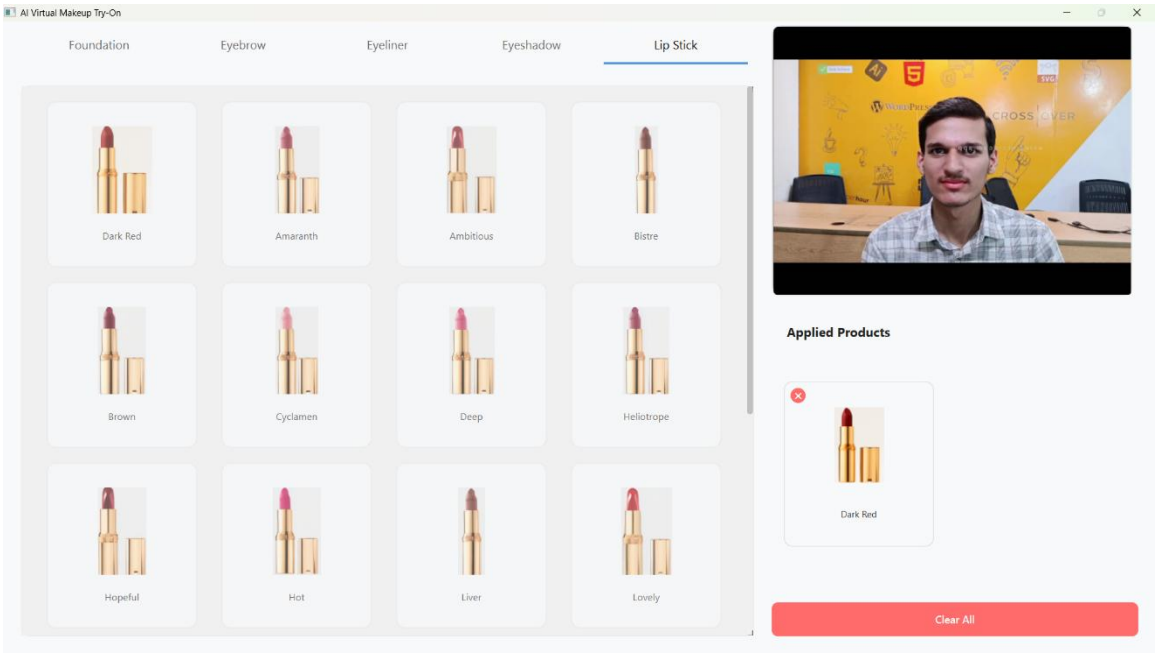


Figure 7.4: Lipstick Product Application Interface

Users can browse and select from vibrant lipstick shades. Here, “Amaranth” is applied to the lips in real-time.

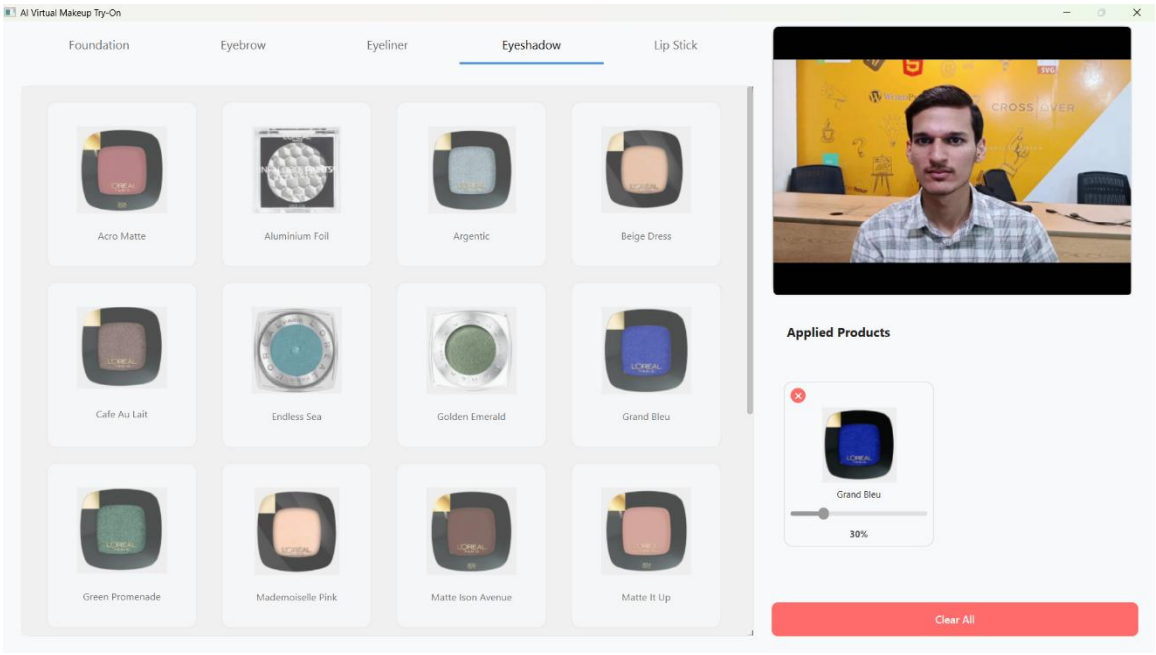


Figure 7.5: Eyeshadow Interface With Opacity Slider

Each eyeshadow can be applied with adjustable intensity, creating realistic gradients. Example shown: “Matte Ison Avenue”.

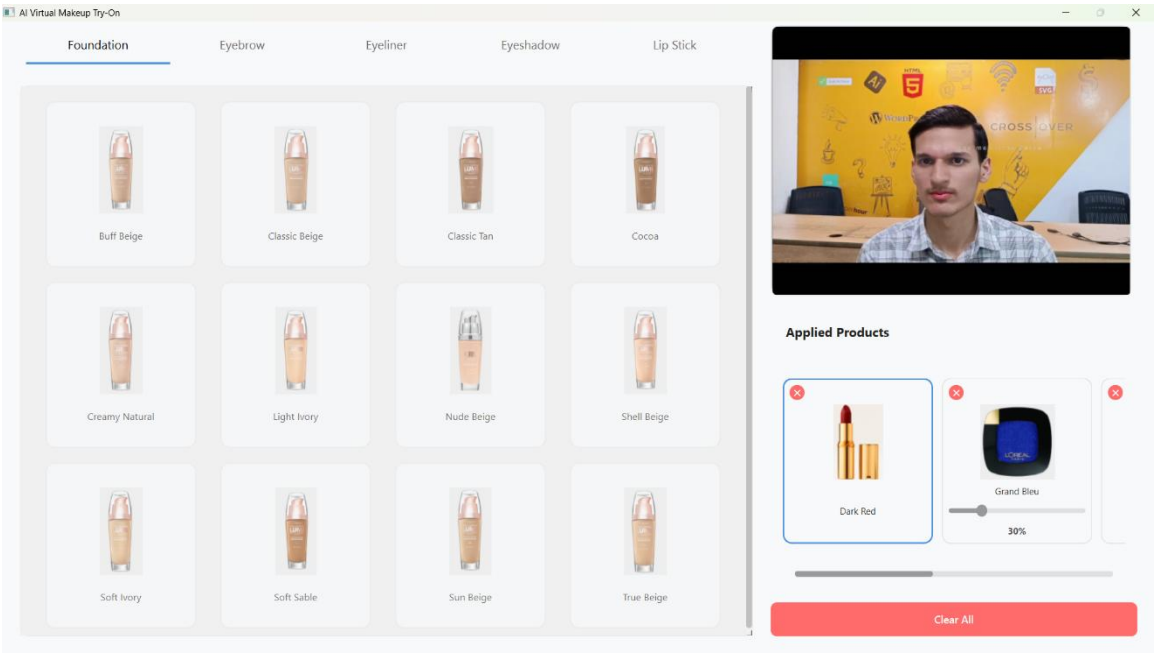


FIGURE 7.6: COMBINED PRODUCT VIEW

The user can apply multiple products simultaneously (foundation + lipstick + eyeliner + eyeshadow + eyebrow), and the "Applied Products" panel reflects the current selections with sliders and remove buttons.

7.5 How to Use the Application

This section provides a complete step-by-step guide for users to operate the AI-Based Virtual Makeup Try-On system from startup to shutdown. The application is designed to be intuitive, even for first-time users with no technical background.

Step 1: Launch the Application

- Run main.py after activating your virtual environment (as described in Section 7.3).
- A GUI window will appear showing the live webcam feed with your face centered in the frame.

Step 2: Select a Makeup Category

At the top of the interface, click on any of the following tabs:

- Foundation

- Eyebrow
- Eyeliner
- Eyeshadow
- Lip Stick

This loads the corresponding set of makeup product options in the grid below.

STEP 3: APPLY A PRODUCT

Click on any product tile (e.g., a lipstick shade).

The selected makeup will instantly appear on your face in the live camera view.

Products are applied using real-time facial landmark tracking provided by MediaPipe [15].

STEP 4: ADJUST PRODUCT INTENSITY

For certain categories like foundation and eyeshadow, an intensity slider appears.

Move the slider left/right to increase or decrease opacity.

This helps simulate a natural or bold look depending on user preference.

STEP 5: SWITCH OR REMOVE PRODUCTS

- To remove a product:
    - Click the “Remove” button beneath the active product.
- To change a product:
    - Simply select a new product from the list; it will replace the previous one.
- Multiple products can be applied simultaneously (e.g., lipstick + eyeshadow + blush).

STEP 6: INTERACT FREELY

- Smile, blink, or move your head left/right.
- The system automatically adjusts the overlay in real-time using facial tracking.
- Performance is optimized to ensure low latency and stable rendering.

STEP 7: EXIT THE APPLICATION

- Click the “X” button or close the window to exit the program.
- The camera feed and GUI will terminate gracefully.

7.6 FEATURES SUMMARY

Feature	Description
Real-Time Camera Feed	Live display using OpenCV for continuous facial tracking
Makeup Product Categories	5 distinct categories: Foundation, Eyebrow, Eyeliner, Eyeshadow, Lipstick
Facial Landmark Tracking	Uses MediaPipe Face Mesh for precise application of virtual makeup
Intensity Adjustment	Sliders for foundation and eyeshadow to fine-tune visibility
Simultaneous Product Application	Supports applying multiple makeup products at once
Dynamic Product Switching	Instantly switch between products without lag or system crash
Modern User Interface	Built with PySide6, featuring organized layout, sliders, and responsive design

Table 7.2: Features Summary

7.7 CHAPTER SUMMARY

This chapter presented a complete overview of the application prototype, including installation requirements, launching procedures, interface design, and user interaction steps. The real-time try-on experience is powered by computer vision and AR techniques, providing users with a seamless and hygienic method to test virtual makeup.

All main characteristics, such as several levels of product categories, facial tracking, and various adjustments of intensities, were well implemented to provide balance between

usability and naturality. The structure of the GUI was also emphasized, and a visual demonstration of proper operations was displayed by means of the user interface screenshots.

Besides proving the technical feasibility of the real-time virtual makeup application, this practical aspect of the project also highlights the increased importance of AI-driven interactive cosmetic activities.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 PROJECT CONCLUSION

The desktop-based system of the FYP developed, conceptualized and assessed the effect of the Final Year Project- AI-Based Virtual Makeup Try-On, which integrates Artificial Intelligence, Computer Vision and Augmented Reality to afford the users a hygienic, easily reachable and interactive means of achieving an instant cosmetic product preview. The system enables users to apply digital versions of various makeup products—such as foundation, eyeliner, eyeshadow, lipstick, and eyebrow color—onto their faces using a standard webcam and without any physical contact.

From inception to final implementation, the primary goal was to replicate the traditional in-store makeup trial experience in a digital and contactless format, particularly addressing the concerns of hygiene, accessibility, and personalization. The solution has to be lightweight, precise, and speedy but with a user friendly interface to the non technical user.

There were several fundamental technologies incorporated into the project design:

- MediaPipe's Face Mesh model for robust real-time facial landmark detection
- OpenCV for camera interfacing and image processing
- PySide6 for modern and responsive graphical user interface design
- ONNX Runtime with GPU support for efficient inference (optional)

Through systematic planning, iterative development using Agile methodology, and continuous feedback incorporation, the system matured from a basic concept to a fully functional and visually stable application prototype. All the features of the cosmetic products were introduced as modular overlays that are dynamically calibrated according to the facial features of the user and appropriate to stay in the same position even when the user is turning or making any of the facial expressions.

This project has always managed to show that the use of AI to virtualize the process of makeup is viable and useful as needed in real-life settings, e.g.:

- Bridal trial simulations
- Salon pre-consultations
- Ecommerce product previews
- Personal cosmetic experimentation at home

In conclusion, the AI-Based Virtual Makeup Try-On system not only met its original functional goals but also offered room for further expansion, both technically and commercially. It can be used as a powerful basis of the future in the digital beauty technology and interactive augmented experiences with consumers.

8.2 PROJECT OUTCOMES

AI-Based Virtual Makeup Try-On project realized some significant attainments on various levels of design, implementation, performance, and usability. Every module was tested thoroughly, interacted with the users and monitored in terms of its performance. The system clearly illustrates the effectiveness of a combination of AI-driven facial tracking and the real-time process of image overlay to create simulations of cosmetic products in a visually and dynamically responsive fashion.

The above consequences are a summary of the prioritable project accomplishments:

8.2.1 REAL-TIME MAKEUP APPLICATION

One of the achievements of the system is that it is able to apply cosmetic products over a live camera with on a real time basis. Lipstick, foundation, eyeliner, the coloring of the eyes and even eyebrows all have makeup overlays that are visually fixed despite the user making slight movements of the face, blinking, having smiles, and tilting the head.

This was possible in the following ways:

- Face Mesh, implemented by MediaPipe, which has an extremely high accuracy and tracks 468 facial landmarks [28]
- OpenCV was used to optimize drawing overlays

- Dynamic blending merging to make sure that there is no jolt in transition between the skin and virtual product

The application has low latency, so there is no awkward experience of using the application thus relevant in services in retailing, cosmetics or custom-made beauty.

8.2.2 Clean, Modern, and User-Friendly Interface

The other notable accomplishment is designing and developing an intuitive and accessible user interface with PySide6. The structure enables the users to:

- Select different makeup product categories via a tabbed interface
- View product thumbnails and select them with a single click
- Adjust the intensity of overlays using interactive sliders (available for foundation and eyeshadow)
- Remove or switch products instantly without restarting the application

The GUI was successfully tested by users with no programming or cosmetic background, and it received positive feedback for its clarity, ease of navigation, and visual coherence.

8.2.3 Accurate Facial Overlay Alignment

The fundamental technical implementation of making the right alignment of virtual makeup on the moving face area was performed effectively. Each category of the products demanded a specific overlay area:

- **Lipstick**: Tracked across the upper and lower lips
- **Eyeliner**: Mapped along the upper eyelid and lash line
- **Eyeshadow**: Blended into the eyelid area and above the crease
- **Foundation**: Covered the full facial mask with gradient blending
- **Eyebrow Coloring**: Precisely aligned with brow contours

Testing confirmed that the overlays stayed anchored to the correct facial regions, even under moderate head movement and lighting variation.

8.2.4 MULTI-PRODUCT LAYER SUPPORT

Users could also use more than one product at a time without lapse in cloud system or overlapping with other products. Such multi-layer capability was essential to realism and matches actual usage of makeup, where people do not apply one product at a time. For example:

- A user might apply a foundation, then the eyeshadow, the eyeliner, and the lipstick each changing the display on its own, but coordinated with the others.

The scalability to be able to bring future type of products like blush, contour, or mascara is also provided in this modular structure.

8.2.5 COMPATIBILITY ACROSS DEVICES

A successful testing of the application was found on:

- A development machine with 8 GB RAM, i7 13th Gen
- A test machine with 16 GB RAM, i7 8th Gen

It performed identically in the two showing compatibility of hardware and strong rendering capabilities on mid to high-end computers. Moreover, it is stand-alone (doesn t rely on the internet) which means that the system can be portable and used offline (in the salon or mobile kiosks, etc).

To conclude, the system has demonstrated to be performance-complete, aesthetically, and learner-centred, fulfilling the goals of the project and creating a convincing argument towards its implementation in the real world or additional academic study.

8.3 CHALLENGES FACED

The AI-Based Virtual Makeup Try-On system development posed variety of technical and design related challenges. Just like any other real-time application that is driven by computer vision, the challenges of having the correct balance between performance, accuracy, and usability needed to be dealt with. This segment profiles the key challenges faced in this project, and the problem solving techniques implemented to fix the problem.

8.3.1 Maintaining Landmark Stability During Facial Movement

One of the most unyielding problems was that virtual makeup overlays were preserved stagnant and in the right position during:

- Head tilts and turns
- Smiling or speaking
- Facial expressions (e.g., blinking, raising eyebrows)

However knowledge on how these problems can be overcome can be attained utilizing the face mesh tracking feature of MediaPipe since the face mesh data was highly dense, light jitter or lags in refreshing of the landmark sometimes made the makeup look floating or mismatched, particularly on areas of movements such as lips and jawline.

Resolution:

- Implemented frame-to-frame smoothing algorithms to reduce jitter.
- Calibrated the overlay positions using average landmark positions over short intervals.
- Improved the coordinate mapping by applying a bounding box buffer for regions with frequent motion (e.g., around the mouth and eyes).

8.3.2 Achieving Realistic Blending and Product Rendering

Simulating realistic makeup requires more than just adding colored masks over facial regions [26]. Products like foundation and eyeshadow needed to:

- Blend gradually with skin tone
- Avoid hard or unnatural edges
- Maintain consistent visual style under changing lighting

This was especially difficult with foundation along the jawline and eyeshadow near the eyelid crease, where face curvature changes abruptly.

Resolution:

- Used alpha transparency and Gaussian blur techniques to blend product edges naturally.
- Calibrated opacity curves to fade products from center to boundary.
- Used grayscale masks to shape products precisely before coloring.

Despite these improvements, further enhancement with 3D surface mapping would be required for professional-grade realism.

8.3.3 Optimizing Application Performance on Mid-Range Hardware

Since the application had to run in real-time, maintaining a high frame rate while tracking 468 facial landmarks and rendering multiple overlays was a performance challenge—particularly on systems with integrated GPUs or limited RAM.

Observed Bottlenecks:

- Live video decoding
- Real-time landmark processing
- Drawing multiple overlays simultaneously

Resolution:

- Used onnxruntime-gpu to offload AI model execution to the GPU where available.
- Limited frame processing to 15–20 FPS, which was visually sufficient while reducing CPU load.
- Optimized overlay rendering by using region-of-interest (ROI) based updates instead of full-frame redraws.

8.3.4 Absence of Facial Diversity in Testing

Due to limited access to a broad user base, testing was primarily conducted on students with similar age ranges and facial structures. This restricted the system's exposure to:

- Varying skin tones

- Different face shapes and sizes
- Uncommon facial features or makeup use cases

Implication:

- Reduced ability to confirm universal landmark accuracy
- Potential misalignment when used by users outside the tested demographics

Future Solution:

- Include larger, annotated datasets representing diverse populations.
- Introduce adaptive landmark calibration during first-time use.

8.3.5 UI Event Conflicts and Overlay Update Timing

Early versions of the system experienced GUI event lags when switching quickly between products or adjusting sliders while overlays were actively rendering. Under certain situations the intensity slider was unresponsive or the application would have caused freezing.

Resolution:

- Separated GUI events from processing loop using Qt signals/slots.
- Implemented debouncing to avoid redundant UI triggers.
- Reduced overlay update frequency to prevent clashes during intense user interaction.

Nevertheless, even despite these difficulties all the critical issues were solved or minimized enough, and did not lead to application crash or unsuccessful operations. The challenges also gave significant insight into real-time system constraints, bottlenecks in user interaction and the complexity of implementing AI in consumer facing interfaces.

8.4 Future Enhancements

Despite the fact that the present form of the AI-Based Virtual Makeup Try-On system is stable, functional, and has been tested and validated, there are a few points that can be

implemented in the future. These improvements are in line towards making the system more realistic, usable, scalable, and commercial ready.

8.4.1 Addition of New Product Categories

The present system facilitates use of foundation, eyeliner, eyeshadow, lipstick and eyebrow coloring. But nowadays cosmetic procedures include a higher level of specialized layers of the products. The application may also be improved in future to include:

- **Blush**: In an effort to achieve natural cheek color coupled with the face anatomy.
- **Contour and Highlighter**: For advanced facial sculpting effects using light and shadow.
- **Mascara**: Applied to eyelashes, requiring precise 3D modeling of lash regions.
- **Contact Lens Simulation**: Changing eye color virtually based on iris detection.
- **Jewelry Preview**: Simulating earrings or nose pins, expanding beyond makeup into accessories.

8.4.2 AI-Driven Skin Tone and Product Recommendation

To personalize the experience further, AI-based modules could be integrated to:

- Automatically detect the user's skin tone.
- Recommend the best-matching makeup shades from a database.
- Suggest complete looks based on current makeup trends or brand collections.

This would transform the system into a smart beauty advisor, not just a try-on tool.

8.4.3 Undo/Redo and History Tracking

A common user expectation in visual applications is the ability to undo or redo changes. Future versions should support:

- Undo to revert the last product or intensity change.
- Redo to reapply a previously removed product.

- History Log showing previously used combinations, shades, or full makeup sets for comparison.

This would enhance usability and promote more experimentation.

8.4.4 Cross-Platform Deployment (Mobile and Web)

To reach a wider audience, the application could be ported to:

- **Android/iOS**: Using cross-platform tools like Flutter or React Native with AI backend integration.
- **WebGL/WebRTC**: For in-browser use without downloads, ideal for ecommerce websites.

This would enable salon customers, online shoppers, and influencers to access the system anywhere.

8.4.5 Enhanced Realism with 3D Face Modeling

Currently, overlays are rendered on 2D facial planes. A future upgrade would involve:

- Mapping overlays to a 3D mesh of the user's face [27].
- Using depth data to align products with facial curvature (e.g., contouring under cheekbones)
- Improving realism of products like blush, mascara, and highlighters, which require depth-aware shading

Technologies such as ARKit (Apple) or DepthAI could be explored for this purpose.

8.5 Final Reflections

This Final Year Project has been a deeply rewarding journey that combined innovation with real-world usability. It challenged the developer to explore cutting-edge tools and frameworks in computer vision, augmented reality, GUI development, and user experience design. More importantly, it demonstrated how artificial intelligence can be meaningfully applied to everyday problems—like the inconvenience of testing makeup—through creative and practical solutions.

Beyond its technical merits, the project introduced key insights into:

- The importance of modular, scalable code design
- The unpredictability of real-time data (camera input, lighting conditions)
- The value of user feedback in improving system design

The successful completion of this system highlights the potential of combining AI and user interface design to create personalized, intelligent experiences that go beyond entertainment to offer practical consumer value. It has laid a solid foundation for future development in digital cosmetics, virtual try-on platforms, and AR-enhanced personal care solutions.

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